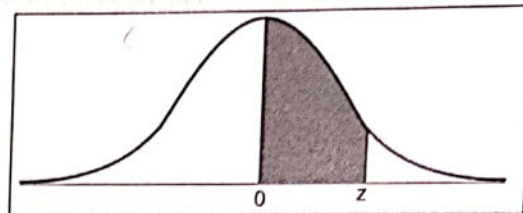


# Areas of a Standard Normal Distribution

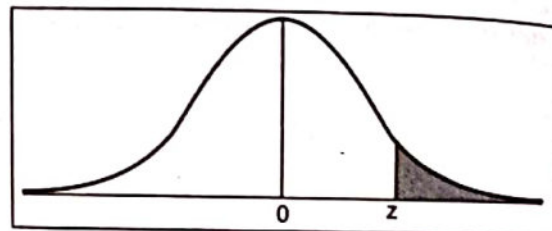


The table entries represent the area under the standard normal curve from 0 to the specified value of  $z$ .

| $z$ | .00   | .01   | .02   | .03   | .04   | .05   | .06   | .07   | .08   | .09   |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0.0 | .0000 | .0040 | .0080 | .0120 | .0160 | .0199 | .0239 | .0279 | .0319 | .0359 |
| 0.1 | .0398 | .0438 | .0478 | .0517 | .0557 | .0596 | .0636 | .0675 | .0714 | .0753 |
| 0.2 | .0793 | .0832 | .0871 | .0910 | .0948 | .0987 | .1026 | .1064 | .1103 | .1141 |
| 0.3 | .1179 | .1217 | .1255 | .1293 | .1331 | .1368 | .1406 | .1443 | .1480 | .1517 |
| 0.4 | .1554 | .1591 | .1628 | .1664 | .1700 | .1736 | .1772 | .1808 | .1844 | .1879 |
| 0.5 | .1915 | .1950 | .1985 | .2019 | .2054 | .2088 | .2123 | .2157 | .2190 | .2224 |
| 0.6 | .2257 | .2291 | .2324 | .2357 | .2389 | .2422 | .2454 | .2486 | .2517 | .2549 |
| 0.7 | .2580 | .2611 | .2642 | .2673 | .2704 | .2734 | .2764 | .2794 | .2823 | .2852 |
| 0.8 | .2881 | .2910 | .2939 | .2967 | .2995 | .3023 | .3051 | .3078 | .3106 | .3133 |
| 0.9 | .3159 | .3186 | .3212 | .3238 | .3264 | .3289 | .3315 | .3340 | .3365 | .3389 |
| 1.0 | .3413 | .3438 | .3461 | .3485 | .3508 | .3531 | .3554 | .3577 | .3599 | .3621 |
| 1.1 | .3643 | .3665 | .3686 | .3708 | .3729 | .3749 | .3770 | .3790 | .3810 | .3830 |
| 1.2 | .3849 | .3869 | .3888 | .3907 | .3925 | .3944 | .3962 | .3980 | .3997 | .4015 |
| 1.3 | .4032 | .4049 | .4066 | .4082 | .4099 | .4115 | .4131 | .4147 | .4162 | .4177 |
| 1.4 | .4192 | .4207 | .4222 | .4236 | .4251 | .4265 | .4279 | .4292 | .4306 | .4319 |
| 1.5 | .4332 | .4345 | .4357 | .4370 | .4382 | .4394 | .4406 | .4418 | .4429 | .4441 |
| 1.6 | .4452 | .4463 | .4474 | .4484 | .4495 | .4505 | .4515 | .4525 | .4535 | .4545 |
| 1.7 | .4554 | .4564 | .4573 | .4582 | .4591 | .4599 | .4608 | .4616 | .4625 | .4633 |
| 1.8 | .4641 | .4649 | .4656 | .4664 | .4671 | .4678 | .4686 | .4693 | .4699 | .4706 |
| 1.9 | .4713 | .4719 | .4726 | .4732 | .4738 | .4744 | .4750 | .4756 | .4761 | .4767 |
| 2.0 | .4772 | .4778 | .4783 | .4788 | .4793 | .4798 | .4803 | .4808 | .4812 | .4817 |
| 2.1 | .4821 | .4826 | .4830 | .4834 | .4838 | .4842 | .4846 | .4850 | .4854 | .4857 |
| 2.2 | .4861 | .4864 | .4868 | .4871 | .4875 | .4878 | .4881 | .4884 | .4887 | .4890 |
| 2.3 | .4893 | .4896 | .4898 | .4901 | .4904 | .4906 | .4909 | .4911 | .4913 | .4916 |
| 2.4 | .4918 | .4920 | .4922 | .4925 | .4927 | .4929 | .4931 | .4932 | .4934 | .4936 |
| 2.5 | .4938 | .4940 | .4941 | .4943 | .4945 | .4946 | .4948 | .4949 | .4951 | .4952 |
| 2.6 | .4953 | .4955 | .4956 | .4957 | .4959 | .4960 | .4961 | .4962 | .4963 | .4964 |
| 2.7 | .4965 | .4966 | .4967 | .4968 | .4969 | .4970 | .4971 | .4972 | .4973 | .4974 |
| 2.8 | .4974 | .4975 | .4976 | .4977 | .4977 | .4978 | .4979 | .4979 | .4980 | .4981 |
| 2.9 | .4981 | .4982 | .4982 | .4983 | .4984 | .4984 | .4985 | .4985 | .4986 | .4986 |
| 3.0 | .4987 | .4987 | .4987 | .4988 | .4988 | .4989 | .4989 | .4989 | .4990 | .4990 |
| 3.1 | .4990 | .4991 | .4991 | .4991 | .4992 | .4992 | .4992 | .4992 | .4993 | .4993 |
| 3.2 | .4993 | .4993 | .4994 | .4994 | .4994 | .4994 | .4994 | .4995 | .4995 | .4995 |
| 3.3 | .4995 | .4995 | .4995 | .4996 | .4996 | .4996 | .4996 | .4996 | .4996 | .4997 |
| 3.4 | .4997 | .4997 | .4997 | .4997 | .4997 | .4997 | .4997 | .4997 | .4997 | .4998 |
| 3.5 | .4998 | .4998 | .4998 | .4998 | .4998 | .4998 | .4998 | .4998 | .4998 | .4998 |
| 3.6 | .4998 | .4998 | .4998 | .4999 | .4999 | .4999 | .4999 | .4999 | .4999 | .4999 |



## Areas of a Standard Normal Distribution (Right)



The table entries represent the area under the standard normal curve more than specified value of  $z$ .

| $z$ | .00    | .01    | .02    | .03    | .04    | .05    | .06    | .07    | .08    | .09    |
|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.0 | 0.5000 | 0.4960 | 0.4920 | 0.4880 | 0.4840 | 0.4801 | 0.4761 | 0.4721 | 0.4681 | 0.4641 |
| 0.1 | 0.4602 | 0.4562 | 0.4522 | 0.4483 | 0.4443 | 0.4404 | 0.4364 | 0.4325 | 0.4286 | 0.4247 |
| 0.2 | 0.4207 | 0.4168 | 0.4129 | 0.4090 | 0.4052 | 0.4013 | 0.3974 | 0.3936 | 0.3897 | 0.3859 |
| 0.3 | 0.3821 | 0.3783 | 0.3745 | 0.3707 | 0.3669 | 0.3632 | 0.3594 | 0.3557 | 0.3520 | 0.3483 |
| 0.4 | 0.3446 | 0.3409 | 0.3372 | 0.3336 | 0.3300 | 0.3264 | 0.3228 | 0.3192 | 0.3156 | 0.3121 |
| 0.5 | 0.3085 | 0.3050 | 0.3015 | 0.2981 | 0.2946 | 0.2912 | 0.2877 | 0.2843 | 0.2810 | 0.2776 |
| 0.6 | 0.2743 | 0.2709 | 0.2676 | 0.2643 | 0.2611 | 0.2578 | 0.2546 | 0.2514 | 0.2483 | 0.2451 |
| 0.7 | 0.2420 | 0.2389 | 0.2358 | 0.2327 | 0.2296 | 0.2266 | 0.2236 | 0.2206 | 0.2177 | 0.2148 |
| 0.8 | 0.2119 | 0.2090 | 0.2061 | 0.2033 | 0.2005 | 0.1977 | 0.1949 | 0.1922 | 0.1894 | 0.1867 |
| 0.9 | 0.1841 | 0.1814 | 0.1788 | 0.1762 | 0.1736 | 0.1711 | 0.1685 | 0.1660 | 0.1635 | 0.1611 |
| 1.0 | 0.1587 | 0.1562 | 0.1539 | 0.1515 | 0.1492 | 0.1469 | 0.1446 | 0.1423 | 0.1401 | 0.1379 |
| 1.1 | 0.1357 | 0.1335 | 0.1314 | 0.1292 | 0.1271 | 0.1251 | 0.1230 | 0.1210 | 0.1190 | 0.1170 |
| 1.2 | 0.1151 | 0.1131 | 0.1112 | 0.1093 | 0.1075 | 0.1056 | 0.1038 | 0.1020 | 0.1003 | 0.0985 |
| 1.3 | 0.0968 | 0.0951 | 0.0934 | 0.0918 | 0.0901 | 0.0885 | 0.0869 | 0.0853 | 0.0838 | 0.0823 |
| 1.4 | 0.0808 | 0.0793 | 0.0778 | 0.0764 | 0.0749 | 0.0735 | 0.0721 | 0.0708 | 0.0694 | 0.0681 |
| 1.5 | 0.0668 | 0.0655 | 0.0643 | 0.0630 | 0.0618 | 0.0606 | 0.0594 | 0.0582 | 0.0571 | 0.0559 |
| 1.6 | 0.0548 | 0.0537 | 0.0526 | 0.0516 | 0.0505 | 0.0495 | 0.0485 | 0.0475 | 0.0465 | 0.0455 |
| 1.7 | 0.0446 | 0.0436 | 0.0427 | 0.0418 | 0.0409 | 0.0401 | 0.0392 | 0.0384 | 0.0375 | 0.0367 |
| 1.8 | 0.0359 | 0.0351 | 0.0344 | 0.0336 | 0.0329 | 0.0322 | 0.0314 | 0.0307 | 0.0301 | 0.0294 |
| 1.9 | 0.0287 | 0.0281 | 0.0274 | 0.0268 | 0.0262 | 0.0256 | 0.0250 | 0.0244 | 0.0239 | 0.0233 |
| 2.0 | 0.0228 | 0.0222 | 0.0217 | 0.0212 | 0.0207 | 0.0202 | 0.0197 | 0.0192 | 0.0188 | 0.0183 |
| 2.1 | 0.0179 | 0.0174 | 0.0170 | 0.0166 | 0.0162 | 0.0158 | 0.0154 | 0.0150 | 0.0146 | 0.0143 |
| 2.2 | 0.0139 | 0.0136 | 0.0132 | 0.0129 | 0.0125 | 0.0122 | 0.0119 | 0.0116 | 0.0113 | 0.0110 |
| 2.3 | 0.0107 | 0.0104 | 0.0102 | 0.0099 | 0.0096 | 0.0094 | 0.0091 | 0.0089 | 0.0087 | 0.0084 |
| 2.4 | 0.0082 | 0.0080 | 0.0078 | 0.0075 | 0.0073 | 0.0071 | 0.0069 | 0.0068 | 0.0066 | 0.0064 |
| 2.5 | 0.0062 | 0.0060 | 0.0059 | 0.0057 | 0.0055 | 0.0054 | 0.0052 | 0.0051 | 0.0049 | 0.0048 |
| 2.6 | 0.0047 | 0.0045 | 0.0044 | 0.0043 | 0.0041 | 0.0040 | 0.0039 | 0.0038 | 0.0037 | 0.0036 |
| 2.7 | 0.0035 | 0.0034 | 0.0033 | 0.0032 | 0.0031 | 0.0030 | 0.0029 | 0.0028 | 0.0027 | 0.0026 |
| 2.8 | 0.0026 | 0.0025 | 0.0024 | 0.0023 | 0.0023 | 0.0022 | 0.0021 | 0.0021 | 0.0020 | 0.0019 |
| 2.9 | 0.0019 | 0.0018 | 0.0018 | 0.0017 | 0.0016 | 0.0016 | 0.0015 | 0.0015 | 0.0014 | 0.0014 |
| 3.0 | 0.0013 | 0.0013 | 0.0013 | 0.0012 | 0.0012 | 0.0011 | 0.0011 | 0.0011 | 0.0010 | 0.0010 |
| 3.1 | 0.0010 | 0.0009 | 0.0009 | 0.0009 | 0.0008 | 0.0008 | 0.0008 | 0.0008 | 0.0007 | 0.0007 |
| 3.2 | 0.0007 | 0.0007 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0005 | 0.0005 |
| 3.3 | 0.0005 | 0.0005 | 0.0005 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0003 |
| 3.4 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0002 |
| 3.5 | 0.0002 | 0.0002 | 0.0002 | 0.0002 | 0.0002 | 0.0002 | 0.0002 | 0.0002 | 0.0002 | 0.0002 |
| 3.6 | 0.0002 | 0.0002 | 0.0002 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.0001 |



**STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.**

| <b>Z</b>   | <b>.00</b> | <b>.01</b> | <b>.02</b> | <b>.03</b> | <b>.04</b> | <b>.05</b> | <b>.06</b> | <b>.07</b> | <b>.08</b> | <b>.09</b> |
|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| <b>0.0</b> | .50000     | .50399     | .50798     | .51197     | .51595     | .51994     | .52392     | .52790     | .53188     | .53586     |
| <b>0.1</b> | .53983     | .54380     | .54776     | .55172     | .55567     | .55962     | .56356     | .56749     | .57142     | .57535     |
| <b>0.2</b> | .57926     | .58317     | .58706     | .59095     | .59483     | .59871     | .60257     | .60642     | .61026     | .61409     |
| <b>0.3</b> | .61791     | .62172     | .62552     | .62930     | .63307     | .63683     | .64058     | .64431     | .64803     | .65173     |
| <b>0.4</b> | .65542     | .65910     | .66276     | .66640     | .67003     | .67364     | .67724     | .68082     | .68439     | .68793     |
| <b>0.5</b> | .69146     | .69497     | .69847     | .70194     | .70540     | .70884     | .71226     | .71566     | .71904     | .72240     |
| <b>0.6</b> | .72575     | .72907     | .73237     | .73565     | .73891     | .74215     | .74537     | .74857     | .75175     | .75490     |
| <b>0.7</b> | .75804     | .76115     | .76424     | .76730     | .77035     | .77337     | .77637     | .77935     | .78230     | .78524     |
| <b>0.8</b> | .78814     | .79103     | .79389     | .79673     | .79955     | .80234     | .80511     | .80785     | .81057     | .81327     |
| <b>0.9</b> | .81594     | .81859     | .82121     | .82381     | .82639     | .82894     | .83147     | .83398     | .83646     | .83891     |
| <b>1.0</b> | .84134     | .84375     | .84614     | .84849     | .85083     | .85314     | .85543     | .85769     | .85993     | .86214     |
| <b>1.1</b> | .86433     | .86650     | .86864     | .87076     | .87286     | .87493     | .87698     | .87900     | .88100     | .88298     |
| <b>1.2</b> | .88493     | .88686     | .88877     | .89065     | .89251     | .89435     | .89617     | .89796     | .89973     | .90147     |
| <b>1.3</b> | .90320     | .90490     | .90658     | .90824     | .90988     | .91149     | .91309     | .91466     | .91621     | .91774     |
| <b>1.4</b> | .91924     | .92073     | .92220     | .92364     | .92507     | .92647     | .92785     | .92922     | .93056     | .93189     |
| <b>1.5</b> | .93319     | .93448     | .93574     | .93699     | .93822     | .93943     | .94062     | .94179     | .94295     | .94408     |
| <b>1.6</b> | .94520     | .94630     | .94738     | .94845     | .94950     | .95053     | .95154     | .95254     | .95352     | .95449     |
| <b>1.7</b> | .95543     | .95637     | .95728     | .95818     | .95907     | .95994     | .96080     | .96164     | .96246     | .96327     |
| <b>1.8</b> | .96407     | .96485     | .96562     | .96638     | .96712     | .96784     | .96856     | .96926     | .96995     | .97062     |
| <b>1.9</b> | .97128     | .97193     | .97257     | .97320     | .97381     | .97441     | .97500     | .97558     | .97615     | .97670     |
| <b>2.0</b> | .97725     | .97778     | .97831     | .97882     | .97932     | .97982     | .98030     | .98077     | .98124     | .98169     |
| <b>2.1</b> | .98214     | .98257     | .98300     | .98341     | .98382     | .98422     | .98461     | .98500     | .98537     | .98574     |
| <b>2.2</b> | .98610     | .98645     | .98679     | .98713     | .98745     | .98778     | .98809     | .98840     | .98870     | .98899     |
| <b>2.3</b> | .98928     | .98956     | .98983     | .99010     | .99036     | .99061     | .99086     | .99111     | .99134     | .99158     |
| <b>2.4</b> | .99180     | .99202     | .99224     | .99245     | .99266     | .99286     | .99305     | .99324     | .99343     | .99361     |
| <b>2.5</b> | .99379     | .99396     | .99413     | .99430     | .99446     | .99461     | .99477     | .99492     | .99506     | .99520     |
| <b>2.6</b> | .99534     | .99547     | .99560     | .99573     | .99585     | .99598     | .99609     | .99621     | .99632     | .99643     |
| <b>2.7</b> | .99653     | .99664     | .99674     | .99683     | .99693     | .99702     | .99711     | .99720     | .99728     | .99736     |
| <b>2.8</b> | .99744     | .99752     | .99760     | .99767     | .99774     | .99781     | .99788     | .99795     | .99801     | .99807     |
| <b>2.9</b> | .99813     | .99819     | .99825     | .99831     | .99836     | .99841     | .99846     | .99851     | .99856     | .99861     |
| <b>3.0</b> | .99865     | .99869     | .99874     | .99878     | .99882     | .99886     | .99889     | .99893     | .99896     | .99900     |
| <b>3.1</b> | .99903     | .99906     | .99910     | .99913     | .99916     | .99918     | .99921     | .99924     | .99926     | .99929     |
| <b>3.2</b> | .99931     | .99934     | .99936     | .99938     | .99940     | .99942     | .99944     | .99946     | .99948     | .99950     |
| <b>3.3</b> | .99952     | .99953     | .99955     | .99957     | .99958     | .99960     | .99961     | .99962     | .99964     | .99965     |
| <b>3.4</b> | .99966     | .99968     | .99969     | .99970     | .99971     | .99972     | .99973     | .99974     | .99975     | .99976     |
| <b>3.5</b> | .99977     | .99978     | .99978     | .99979     | .99980     | .99981     | .99981     | .99982     | .99983     | .99983     |
| <b>3.6</b> | .99984     | .99985     | .99985     | .99986     | .99986     | .99987     | .99987     | .99988     | .99988     | .99989     |
| <b>3.7</b> | .99989     | .99990     | .99990     | .99990     | .99991     | .99991     | .99992     | .99992     | .99992     | .99992     |
| <b>3.8</b> | .99993     | .99993     | .99993     | .99994     | .99994     | .99994     | .99994     | .99995     | .99995     | .99995     |
| <b>3.9</b> | .99995     | .99995     | .99996     | .99996     | .99996     | .99996     | .99996     | .99996     | .99997     | .99997     |

| z    | 0.00  | 0.01  | 0.02  | 0.03  | 0.04  | 0.05  | 0.06  | 0.07  | 0.08  | 0.09 |
|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
| -3.6 | .0002 | .0002 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .000 |
| -3.5 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 | .000 |
| -3.4 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .000 |
| -3.3 | .0005 | .0005 | .0005 | .0004 | .0004 | .0004 | .0004 | .0004 | .0004 | .000 |
| -3.2 | .0007 | .0007 | .0006 | .0006 | .0006 | .0006 | .0006 | .0005 | .0005 | .000 |
| -3.1 | .0010 | .0009 | .0009 | .0009 | .0008 | .0008 | .0008 | .0008 | .0007 | .000 |
| -3.0 | .0013 | .0013 | .0013 | .0012 | .0012 | .0011 | .0011 | .0011 | .0010 | .001 |
| -2.9 | .0019 | .0018 | .0018 | .0017 | .0016 | .0016 | .0015 | .0015 | .0014 | .001 |
| -2.8 | .0026 | .0025 | .0024 | .0023 | .0023 | .0022 | .0021 | .0021 | .0020 | .001 |
| -2.7 | .0035 | .0034 | .0033 | .0032 | .0031 | .0030 | .0029 | .0028 | .0027 | .002 |
| -2.6 | .0047 | .0045 | .0044 | .0043 | .0041 | .0040 | .0039 | .0038 | .0037 | .003 |
| -2.5 | .0062 | .0060 | .0059 | .0057 | .0055 | .0054 | .0052 | .0051 | .0049 | .004 |
| -2.4 | .0082 | .0080 | .0078 | .0075 | .0073 | .0071 | .0069 | .0068 | .0066 | .006 |
| -2.3 | .0107 | .0104 | .0102 | .0099 | .0096 | .0094 | .0091 | .0089 | .0087 | .008 |
| -2.2 | .0139 | .0136 | .0132 | .0129 | .0125 | .0122 | .0119 | .0116 | .0113 | .011 |
| -2.1 | .0179 | .0174 | .0170 | .0166 | .0162 | .0158 | .0154 | .0150 | .0146 | .014 |
| -2.0 | .0228 | .0222 | .0217 | .0212 | .0207 | .0202 | .0197 | .0192 | .0188 | .018 |
| -1.9 | .0287 | .0281 | .0274 | .0268 | .0262 | .0256 | .0250 | .0244 | .0239 | .023 |
| -1.8 | .0359 | .0351 | .0344 | .0336 | .0329 | .0322 | .0314 | .0307 | .0301 | .029 |
| -1.7 | .0446 | .0436 | .0427 | .0418 | .0409 | .0401 | .0392 | .0384 | .0375 | .036 |
| -1.6 | .0548 | .0537 | .0526 | .0516 | .0505 | .0495 | .0485 | .0475 | .0465 | .045 |
| -1.5 | .0668 | .0655 | .0643 | .0630 | .0618 | .0606 | .0594 | .0582 | .0571 | .055 |
| -1.4 | .0808 | .0793 | .0778 | .0764 | .0749 | .0735 | .0721 | .0708 | .0694 | .068 |
| -1.3 | .0968 | .0951 | .0934 | .0918 | .0901 | .0885 | .0869 | .0853 | .0838 | .082 |
| -1.2 | .1151 | .1131 | .1112 | .1093 | .1075 | .1056 | .1038 | .1020 | .1003 | .098 |
| -1.1 | .1357 | .1335 | .1314 | .1292 | .1271 | .1251 | .1230 | .1210 | .1190 | .117 |
| -1.0 | .1587 | .1562 | .1539 | .1515 | .1492 | .1469 | .1446 | .1423 | .1401 | .137 |
| -0.9 | .1841 | .1814 | .1788 | .1762 | .1736 | .1711 | .1685 | .1660 | .1635 | .161 |
| -0.8 | .2119 | .2090 | .2061 | .2033 | .2005 | .1977 | .1949 | .1922 | .1894 | .186 |
| -0.7 | .2420 | .2389 | .2358 | .2327 | .2296 | .2266 | .2236 | .2206 | .2177 | .214 |
| -0.6 | .2743 | .2709 | .2676 | .2643 | .2611 | .2578 | .2546 | .2514 | .2483 | .245 |
| -0.5 | .3085 | .3050 | .3015 | .2981 | .2946 | .2912 | .2877 | .2843 | .2810 | .277 |
| -0.4 | .3446 | .3409 | .3372 | .3336 | .3300 | .3264 | .3228 | .3192 | .3156 | .312 |
| -0.3 | .3821 | .3783 | .3745 | .3707 | .3669 | .3632 | .3594 | .3557 | .3520 | .348 |
| -0.2 | .4207 | .4168 | .4129 | .4090 | .4052 | .4013 | .3974 | .3936 | .3897 | .385 |
| -0.1 | .4602 | .4562 | .4522 | .4483 | .4443 | .4404 | .4364 | .4325 | .4286 | .424 |



Critical Values of Student's t-distribution

| d.f. | Level of significance for one-tailed test |       |        |        |        |         |
|------|-------------------------------------------|-------|--------|--------|--------|---------|
|      | 0.10                                      | 0.05  | 0.025  | 0.01   | 0.005  | 0.0005  |
|      | Level of significance for two-tailed test |       |        |        |        |         |
|      | 0.20                                      | 0.10  | 0.05   | 0.02   | 0.01   | 0.001   |
| 1    | 3.078                                     | 6.314 | 12.706 | 31.821 | 63.657 | 636.619 |
| 2    | 1.886                                     | 2.920 | 4.303  | 6.965  | 9.925  | 31.599  |
| 3    | 1.638                                     | 2.353 | 3.182  | 4.541  | 5.841  | 12.924  |
| 4    | 1.533                                     | 2.132 | 2.776  | 3.747  | 4.604  | 8.610   |
| 5    | 1.476                                     | 2.015 | 2.571  | 3.365  | 4.032  | 6.869   |
| 6    | 1.440                                     | 1.943 | 2.447  | 3.143  | 3.707  | 5.959   |
| 7    | 1.415                                     | 1.895 | 2.365  | 2.998  | 3.499  | 5.408   |
| 8    | 1.397                                     | 1.860 | 2.306  | 2.896  | 3.355  | 5.041   |
| 9    | 1.383                                     | 1.833 | 2.262  | 2.821  | 3.250  | 4.781   |
| 10   | 1.372                                     | 1.812 | 2.228  | 2.764  | 3.169  | 4.587   |
| 11   | 1.363                                     | 1.796 | 2.201  | 2.718  | 3.106  | 4.437   |
| 12   | 1.356                                     | 1.782 | 2.179  | 2.681  | 3.055  | 4.318   |
| 13   | 1.350                                     | 1.771 | 2.160  | 2.650  | 3.012  | 4.221   |
| 14   | 1.345                                     | 1.761 | 2.145  | 2.624  | 2.977  | 4.140   |
| 15   | 1.341                                     | 1.753 | 2.131  | 2.602  | 2.947  | 4.073   |
| 16   | 1.337                                     | 1.746 | 2.120  | 2.583  | 2.921  | 4.015   |
| 17   | 1.333                                     | 1.740 | 2.110  | 2.567  | 2.898  | 3.965   |
| 18   | 1.330                                     | 1.734 | 2.101  | 2.552  | 2.878  | 3.922   |
| 19   | 1.328                                     | 1.729 | 2.093  | 2.539  | 2.861  | 3.883   |
| 20   | 1.325                                     | 1.725 | 2.086  | 2.528  | 2.845  | 3.850   |
| 21   | 1.323                                     | 1.721 | 2.080  | 2.518  | 2.831  | 3.819   |
| 22   | 1.321                                     | 1.717 | 2.074  | 2.508  | 2.819  | 3.792   |
| 23   | 1.319                                     | 1.714 | 2.069  | 2.500  | 2.807  | 3.768   |
| 24   | 1.318                                     | 1.711 | 2.064  | 2.492  | 2.797  | 3.745   |
| 25   | 1.316                                     | 1.708 | 2.060  | 2.485  | 2.787  | 3.725   |
| 26   | 1.315                                     | 1.706 | 2.056  | 2.479  | 2.779  | 3.707   |
| 27   | 1.314                                     | 1.703 | 2.052  | 2.473  | 2.771  | 3.690   |
| 28   | 1.313                                     | 1.701 | 2.048  | 2.467  | 2.763  | 3.674   |
| 29   | 1.311                                     | 1.699 | 2.045  | 2.462  | 2.756  | 3.659   |
| 30   | 1.310                                     | 1.697 | 2.042  | 2.457  | 2.750  | 3.646   |
| 40   | 1.303                                     | 1.684 | 2.021  | 2.423  | 2.704  | 3.551   |
| 60   | 1.296                                     | 1.671 | 2.000  | 2.390  | 2.660  | 3.460   |
| 120  | 1.289                                     | 1.658 | 1.980  | 2.358  | 2.617  | 3.373   |
| ∞    | 1.282                                     | 1.645 | 1.960  | 2.326  | 2.576  | 3.291   |



## Critical Values of Chi-Square

The values of  $\chi^2$  correspond to a specific right-tail area and specific number of degrees of freedom df.

| Degree of Freedom | Level of Significance |         |         |         |         |         |
|-------------------|-----------------------|---------|---------|---------|---------|---------|
|                   | 0.2                   | 0.1     | 0.05    | 0.02    | 0.01    | 0.001   |
| 1                 | 1.642                 | 2.706   | 3.841   | 5.412   | 6.635   | 10.828  |
| 2                 | 3.219                 | 4.605   | 5.991   | 7.824   | 9.210   | 13.816  |
| 3                 | 4.642                 | 6.251   | 7.815   | 9.837   | 11.345  | 16.266  |
| 4                 | 5.989                 | 7.779   | 9.488   | 11.668  | 13.277  | 18.467  |
| 5                 | 7.289                 | 9.236   | 11.070  | 13.388  | 15.086  | 20.515  |
| 6                 | 8.558                 | 10.645  | 12.592  | 15.033  | 16.812  | 22.458  |
| 7                 | 9.803                 | 12.017  | 14.067  | 16.622  | 18.475  | 24.322  |
| 8                 | 11.030                | 13.362  | 15.507  | 18.168  | 20.090  | 26.124  |
| 9                 | 12.242                | 14.684  | 16.919  | 19.679  | 21.666  | 27.877  |
| 10                | 13.442                | 15.987  | 18.307  | 21.161  | 23.209  | 29.588  |
| 11                | 14.631                | 17.275  | 19.675  | 22.618  | 24.725  | 31.264  |
| 12                | 15.812                | 18.549  | 21.026  | 24.054  | 26.217  | 32.909  |
| 13                | 16.985                | 19.812  | 22.362  | 25.472  | 27.688  | 34.528  |
| 14                | 18.151                | 21.064  | 23.685  | 26.873  | 29.141  | 36.123  |
| 15                | 19.311                | 22.307  | 24.996  | 28.259  | 30.578  | 37.697  |
| 16                | 20.465                | 23.542  | 26.296  | 29.633  | 32.000  | 39.252  |
| 17                | 21.615                | 24.769  | 27.587  | 30.995  | 33.409  | 40.790  |
| 18                | 22.760                | 25.989  | 28.869  | 32.346  | 34.805  | 42.312  |
| 19                | 23.900                | 27.204  | 30.144  | 33.687  | 36.191  | 43.820  |
| 20                | 25.038                | 28.412  | 31.410  | 35.020  | 37.566  | 45.315  |
| 21                | 26.171                | 29.615  | 32.671  | 36.343  | 38.932  | 46.797  |
| 22                | 27.301                | 30.813  | 33.924  | 37.659  | 40.289  | 48.268  |
| 23                | 28.429                | 32.007  | 35.172  | 38.968  | 41.638  | 49.728  |
| 24                | 29.553                | 33.196  | 36.415  | 40.270  | 42.980  | 51.179  |
| 25                | 30.675                | 34.382  | 37.652  | 41.566  | 44.314  | 52.620  |
| 26                | 31.795                | 35.563  | 38.885  | 42.856  | 45.642  | 54.052  |
| 27                | 32.912                | 36.741  | 40.113  | 44.140  | 46.963  | 55.476  |
| 28                | 34.027                | 37.916  | 41.337  | 45.419  | 48.278  | 56.892  |
| 29                | 35.139                | 39.087  | 42.557  | 46.693  | 49.588  | 58.301  |
| 30                | 36.250                | 40.256  | 43.773  | 47.962  | 50.892  | 59.703  |
| 40                | 47.269                | 51.805  | 55.758  | 60.436  | 63.691  | 73.402  |
| 60                | 68.972                | 74.397  | 79.082  | 84.580  | 88.379  | 99.607  |
| 120               | 132.806               | 140.233 | 146.567 | 153.918 | 158.950 | 173.617 |



F Table for  $\alpha = 0.10$ 

Degree of Freedom for Numerator

| df <sub>1</sub> \ df <sub>2</sub> | 1      | 2      | 3      | 4      | 5      | 6      | 7      | 8      | 9      | 10     | 12     | 15     | 20     | 24     | 30     | 40     | 60     | 120    | $\infty$ |
|-----------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|----------|
| 1                                 | 39.863 | 49.500 | 53.593 | 55.832 | 57.240 | 58.204 | 58.905 | 59.438 | 59.857 | 60.194 | 60.705 | 61.220 | 61.740 | 62.002 | 62.264 | 62.529 | 62.794 | 63.060 | 63.323   |
| 2                                 | 8.526  | 9.000  | 9.161  | 9.243  | 9.292  | 9.325  | 9.349  | 9.366  | 9.380  | 9.391  | 9.409  | 9.424  | 9.441  | 9.449  | 9.457  | 9.466  | 9.474  | 9.482  | 9.491    |
| 3                                 | 5.538  | 5.462  | 5.390  | 5.342  | 5.309  | 5.284  | 5.268  | 5.251  | 5.240  | 5.230  | 5.215  | 5.200  | 5.184  | 5.176  | 5.168  | 5.159  | 5.151  | 5.142  | 5.133    |
| 4                                 | 4.544  | 4.324  | 4.190  | 4.107  | 4.050  | 4.009  | 3.978  | 3.954  | 3.935  | 3.919  | 3.895  | 3.870  | 3.844  | 3.830  | 3.817  | 3.803  | 3.789  | 3.775  | 3.760    |
| 5                                 | 4.060  | 3.779  | 3.619  | 3.520  | 3.452  | 3.404  | 3.367  | 3.339  | 3.316  | 3.297  | 3.268  | 3.238  | 3.206  | 3.190  | 3.174  | 3.157  | 3.140  | 3.122  | 3.105    |
| 6                                 | 3.775  | 3.463  | 3.288  | 3.180  | 3.107  | 3.054  | 3.014  | 2.983  | 2.957  | 2.936  | 2.904  | 2.871  | 2.836  | 2.818  | 2.799  | 2.781  | 2.761  | 2.742  | 2.722    |
| 7                                 | 3.589  | 3.257  | 3.074  | 2.960  | 2.883  | 2.827  | 2.784  | 2.751  | 2.724  | 2.702  | 2.668  | 2.632  | 2.594  | 2.575  | 2.555  | 2.535  | 2.514  | 2.492  | 2.470    |
| 8                                 | 3.457  | 3.113  | 2.923  | 2.805  | 2.726  | 2.668  | 2.624  | 2.589  | 2.561  | 2.538  | 2.501  | 2.464  | 2.424  | 2.404  | 2.383  | 2.361  | 2.339  | 2.316  | 2.292    |
| 9                                 | 3.360  | 3.006  | 2.812  | 2.692  | 2.610  | 2.550  | 2.505  | 2.469  | 2.440  | 2.416  | 2.378  | 2.339  | 2.298  | 2.276  | 2.254  | 2.231  | 2.208  | 2.184  | 2.159    |
| 10                                | 3.285  | 2.924  | 2.727  | 2.605  | 2.521  | 2.460  | 2.413  | 2.377  | 2.347  | 2.322  | 2.284  | 2.243  | 2.200  | 2.178  | 2.155  | 2.131  | 2.107  | 2.081  | 2.055    |
| 11                                | 3.225  | 2.859  | 2.660  | 2.536  | 2.451  | 2.389  | 2.341  | 2.304  | 2.273  | 2.248  | 2.209  | 2.167  | 2.123  | 2.100  | 2.076  | 2.051  | 2.026  | 1.999  | 1.972    |
| 12                                | 3.176  | 2.806  | 2.605  | 2.480  | 2.394  | 2.331  | 2.282  | 2.244  | 2.213  | 2.187  | 2.147  | 2.104  | 2.059  | 2.035  | 2.011  | 1.986  | 1.959  | 1.932  | 1.903    |
| 13                                | 3.136  | 2.763  | 2.560  | 2.433  | 2.346  | 2.282  | 2.234  | 2.195  | 2.163  | 2.137  | 2.096  | 2.053  | 2.008  | 1.982  | 1.957  | 1.931  | 1.904  | 1.875  | 1.845    |
| 14                                | 3.102  | 2.726  | 2.522  | 2.394  | 2.306  | 2.242  | 2.193  | 2.153  | 2.121  | 2.095  | 2.053  | 2.009  | 1.962  | 1.937  | 1.911  | 1.885  | 1.857  | 1.828  | 1.797    |
| 15                                | 3.073  | 2.695  | 2.489  | 2.361  | 2.273  | 2.208  | 2.158  | 2.118  | 2.086  | 2.060  | 2.017  | 1.972  | 1.924  | 1.899  | 1.872  | 1.845  | 1.816  | 1.786  | 1.755    |
| 16                                | 3.048  | 2.668  | 2.461  | 2.332  | 2.243  | 2.178  | 2.128  | 2.087  | 2.055  | 2.028  | 1.985  | 1.939  | 1.891  | 1.865  | 1.838  | 1.810  | 1.781  | 1.750  | 1.718    |
| 17                                | 3.026  | 2.644  | 2.437  | 2.307  | 2.218  | 2.152  | 2.101  | 2.061  | 2.028  | 1.999  | 1.957  | 1.911  | 1.862  | 1.836  | 1.809  | 1.780  | 1.750  | 1.719  | 1.685    |
| 18                                | 3.006  | 2.623  | 2.416  | 2.285  | 2.195  | 2.129  | 2.078  | 2.037  | 2.004  | 1.976  | 1.933  | 1.886  | 1.836  | 1.810  | 1.782  | 1.753  | 1.723  | 1.690  | 1.656    |
| 19                                | 2.989  | 2.605  | 2.397  | 2.266  | 2.175  | 2.109  | 2.058  | 2.017  | 1.983  | 1.955  | 1.911  | 1.864  | 1.814  | 1.787  | 1.759  | 1.729  | 1.698  | 1.665  | 1.630    |
| 20                                | 2.974  | 2.589  | 2.380  | 2.248  | 2.158  | 2.091  | 2.039  | 1.998  | 1.964  | 1.936  | 1.892  | 1.844  | 1.793  | 1.766  | 1.738  | 1.708  | 1.676  | 1.643  | 1.607    |
| 21                                | 2.960  | 2.574  | 2.364  | 2.233  | 2.142  | 2.075  | 2.023  | 1.981  | 1.947  | 1.919  | 1.874  | 1.827  | 1.775  | 1.748  | 1.719  | 1.688  | 1.656  | 1.622  | 1.586    |
| 22                                | 2.948  | 2.561  | 2.351  | 2.219  | 2.127  | 2.060  | 2.008  | 1.966  | 1.932  | 1.904  | 1.859  | 1.811  | 1.758  | 1.731  | 1.702  | 1.671  | 1.638  | 1.604  | 1.566    |
| 23                                | 2.937  | 2.549  | 2.338  | 2.206  | 2.114  | 2.047  | 1.994  | 1.953  | 1.918  | 1.890  | 1.844  | 1.795  | 1.743  | 1.715  | 1.686  | 1.655  | 1.622  | 1.587  | 1.549    |
| 24                                | 2.927  | 2.538  | 2.327  | 2.194  | 2.103  | 2.035  | 1.982  | 1.940  | 1.906  | 1.877  | 1.831  | 1.783  | 1.730  | 1.701  | 1.672  | 1.640  | 1.607  | 1.571  | 1.532    |
| 25                                | 2.917  | 2.528  | 2.317  | 2.184  | 2.092  | 2.024  | 1.971  | 1.929  | 1.894  | 1.865  | 1.820  | 1.770  | 1.717  | 1.688  | 1.658  | 1.627  | 1.593  | 1.557  | 1.517    |
| 26                                | 2.909  | 2.519  | 2.307  | 2.174  | 2.082  | 2.013  | 1.961  | 1.918  | 1.884  | 1.855  | 1.809  | 1.759  | 1.705  | 1.677  | 1.646  | 1.614  | 1.580  | 1.543  | 1.503    |
| 27                                | 2.901  | 2.510  | 2.298  | 2.165  | 2.072  | 2.004  | 1.951  | 1.909  | 1.874  | 1.845  | 1.798  | 1.749  | 1.695  | 1.666  | 1.635  | 1.603  | 1.568  | 1.531  | 1.490    |
| 28                                | 2.893  | 2.502  | 2.290  | 2.157  | 2.064  | 1.995  | 1.942  | 1.900  | 1.865  | 1.835  | 1.789  | 1.739  | 1.685  | 1.656  | 1.625  | 1.592  | 1.557  | 1.519  | 1.478    |
| 29                                | 2.887  | 2.495  | 2.283  | 2.149  | 2.056  | 1.987  | 1.934  | 1.891  | 1.856  | 1.827  | 1.780  | 1.730  | 1.675  | 1.646  | 1.615  | 1.582  | 1.547  | 1.508  | 1.467    |
| 30                                | 2.880  | 2.488  | 2.276  | 2.142  | 2.049  | 1.980  | 1.926  | 1.884  | 1.848  | 1.819  | 1.772  | 1.722  | 1.667  | 1.637  | 1.606  | 1.573  | 1.537  | 1.498  | 1.456    |
| 40                                | 2.835  | 2.440  | 2.226  | 2.090  | 1.996  | 1.926  | 1.872  | 1.828  | 1.792  | 1.762  | 1.714  | 1.662  | 1.605  | 1.574  | 1.541  | 1.505  | 1.467  | 1.428  | 1.376    |
| 60                                | 2.791  | 2.393  | 2.177  | 2.040  | 1.945  | 1.874  | 1.819  | 1.774  | 1.738  | 1.707  | 1.657  | 1.603  | 1.543  | 1.510  | 1.475  | 1.437  | 1.395  | 1.347  | 1.291    |
| 120                               | 2.747  | 2.347  | 2.129  | 1.992  | 1.895  | 1.823  | 1.767  | 1.721  | 1.684  | 1.652  | 1.601  | 1.545  | 1.482  | 1.447  | 1.409  | 1.367  | 1.320  | 1.264  | 1.192    |
| $\infty$                          | 2.705  | 2.302  | 2.083  | 1.944  | 1.847  | 1.774  | 1.716  | 1.670  | 1.631  | 1.598  | 1.545  | 1.487  | 1.420  | 1.383  | 1.341  | 1.295  | 1.239  | 1.168  | 1.000    |

Degree of Freedom for Denominator



F Table for  $\alpha = 0.05$ 

Degree of Freedom for Numerator

| df <sub>1</sub> \ df <sub>2</sub> | 1       | 2       | 3       | 4       | 5       | 6       | 7       | 8       | 9       | 10      | 12      | 15      | 20      | 24      | 30      | 40      | 60      | 120     | $\infty$ |
|-----------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|----------|
| 1                                 | 161.447 | 199.500 | 215.707 | 224.583 | 230.161 | 233.986 | 236.768 | 238.882 | 240.543 | 241.881 | 243.906 | 245.949 | 248.013 | 249.051 | 250.095 | 251.143 | 252.195 | 253.252 | 254.314  |
| 2                                 | 18.512  | 19.000  | 19.164  | 19.246  | 19.296  | 19.329  | 19.353  | 19.371  | 19.384  | 19.395  | 19.412  | 19.429  | 19.445  | 19.454  | 19.462  | 19.470  | 19.479  | 19.487  | 19.495   |
| 3                                 | 10.128  | 9.552   | 9.276   | 9.117   | 9.013   | 8.940   | 8.886   | 8.845   | 8.812   | 8.785   | 8.744   | 8.702   | 8.660   | 8.638   | 8.616   | 8.594   | 8.572   | 8.549   | 8.526    |
| 4                                 | 7.708   | 6.944   | 6.591   | 6.388   | 6.256   | 6.163   | 6.094   | 6.041   | 5.998   | 5.964   | 5.911   | 5.857   | 5.802   | 5.774   | 5.745   | 5.717   | 5.687   | 5.658   | 5.628    |
| 5                                 | 6.607   | 5.786   | 5.409   | 5.192   | 5.050   | 4.950   | 4.875   | 4.818   | 4.772   | 4.735   | 4.677   | 4.618   | 4.558   | 4.527   | 4.495   | 4.463   | 4.431   | 4.398   | 4.365    |
| 6                                 | 5.987   | 5.143   | 4.757   | 4.533   | 4.387   | 4.283   | 4.206   | 4.146   | 4.099   | 4.060   | 3.999   | 3.938   | 3.874   | 3.841   | 3.808   | 3.774   | 3.739   | 3.704   | 3.668    |
| 7                                 | 5.591   | 4.737   | 4.346   | 4.120   | 3.971   | 3.866   | 3.787   | 3.725   | 3.676   | 3.635   | 3.574   | 3.510   | 3.444   | 3.410   | 3.375   | 3.340   | 3.304   | 3.267   | 3.229    |
| 8                                 | 5.317   | 4.459   | 4.066   | 3.837   | 3.687   | 3.580   | 3.500   | 3.438   | 3.388   | 3.347   | 3.283   | 3.218   | 3.150   | 3.115   | 3.079   | 3.042   | 3.005   | 2.968   | 2.927    |
| 9                                 | 5.117   | 4.256   | 3.862   | 3.633   | 3.481   | 3.373   | 3.292   | 3.229   | 3.178   | 3.137   | 3.072   | 3.006   | 2.936   | 2.900   | 2.863   | 2.825   | 2.787   | 2.747   | 2.706    |
| 10                                | 4.964   | 4.102   | 3.708   | 3.478   | 3.325   | 3.217   | 3.135   | 3.071   | 3.020   | 2.978   | 2.913   | 2.845   | 2.774   | 2.737   | 2.699   | 2.660   | 2.621   | 2.580   | 2.537    |
| 11                                | 4.844   | 3.982   | 3.587   | 3.358   | 3.203   | 3.094   | 3.012   | 2.948   | 2.896   | 2.853   | 2.787   | 2.718   | 2.646   | 2.609   | 2.570   | 2.530   | 2.490   | 2.448   | 2.404    |
| 12                                | 4.747   | 3.885   | 3.490   | 3.259   | 3.105   | 2.995   | 2.913   | 2.848   | 2.796   | 2.753   | 2.686   | 2.616   | 2.543   | 2.505   | 2.466   | 2.425   | 2.384   | 2.341   | 2.296    |
| 13                                | 4.667   | 3.805   | 3.410   | 3.179   | 3.025   | 2.915   | 2.832   | 2.766   | 2.714   | 2.671   | 2.603   | 2.533   | 2.458   | 2.420   | 2.380   | 2.339   | 2.296   | 2.252   | 2.206    |
| 14                                | 4.600   | 3.738   | 3.343   | 3.112   | 2.958   | 2.847   | 2.764   | 2.698   | 2.645   | 2.602   | 2.534   | 2.463   | 2.387   | 2.348   | 2.308   | 2.266   | 2.222   | 2.177   | 2.130    |
| 15                                | 4.543   | 3.682   | 3.287   | 3.055   | 2.901   | 2.790   | 2.706   | 2.640   | 2.587   | 2.543   | 2.475   | 2.403   | 2.327   | 2.287   | 2.246   | 2.204   | 2.160   | 2.114   | 2.065    |
| 16                                | 4.494   | 3.633   | 3.238   | 3.006   | 2.852   | 2.741   | 2.657   | 2.591   | 2.537   | 2.493   | 2.424   | 2.352   | 2.275   | 2.235   | 2.193   | 2.150   | 2.105   | 2.058   | 2.009    |
| 17                                | 4.451   | 3.591   | 3.196   | 2.964   | 2.810   | 2.698   | 2.614   | 2.548   | 2.494   | 2.449   | 2.380   | 2.307   | 2.230   | 2.189   | 2.147   | 2.104   | 2.058   | 2.010   | 1.960    |
| 18                                | 4.413   | 3.554   | 3.159   | 2.927   | 2.772   | 2.661   | 2.576   | 2.510   | 2.456   | 2.411   | 2.342   | 2.268   | 2.190   | 2.149   | 2.107   | 2.062   | 2.016   | 1.968   | 1.916    |
| 19                                | 4.380   | 3.521   | 3.127   | 2.895   | 2.740   | 2.628   | 2.543   | 2.476   | 2.422   | 2.377   | 2.308   | 2.234   | 2.155   | 2.114   | 2.071   | 2.026   | 1.979   | 1.930   | 1.878    |
| 20                                | 4.351   | 3.492   | 3.098   | 2.866   | 2.710   | 2.599   | 2.514   | 2.447   | 2.392   | 2.347   | 2.277   | 2.203   | 2.124   | 2.082   | 2.039   | 1.993   | 1.946   | 1.896   | 1.843    |
| 21                                | 4.324   | 3.466   | 3.072   | 2.840   | 2.684   | 2.572   | 2.487   | 2.420   | 2.365   | 2.321   | 2.250   | 2.175   | 2.096   | 2.054   | 2.010   | 1.964   | 1.916   | 1.865   | 1.811    |
| 22                                | 4.300   | 3.443   | 3.049   | 2.816   | 2.661   | 2.549   | 2.463   | 2.396   | 2.341   | 2.296   | 2.225   | 2.150   | 2.070   | 2.028   | 1.984   | 1.938   | 1.889   | 1.838   | 1.783    |
| 23                                | 4.279   | 3.422   | 3.028   | 2.795   | 2.640   | 2.527   | 2.442   | 2.374   | 2.320   | 2.274   | 2.203   | 2.128   | 2.047   | 2.005   | 1.960   | 1.913   | 1.864   | 1.812   | 1.757    |
| 24                                | 4.259   | 3.402   | 3.008   | 2.776   | 2.620   | 2.508   | 2.422   | 2.355   | 2.299   | 2.253   | 2.183   | 2.107   | 2.026   | 1.983   | 1.939   | 1.892   | 1.842   | 1.789   | 1.733    |
| 25                                | 4.241   | 3.385   | 2.991   | 2.758   | 2.603   | 2.490   | 2.404   | 2.337   | 2.282   | 2.236   | 2.164   | 2.088   | 2.007   | 1.964   | 1.919   | 1.871   | 1.821   | 1.768   | 1.711    |
| 26                                | 4.225   | 3.369   | 2.975   | 2.742   | 2.586   | 2.474   | 2.388   | 2.320   | 2.265   | 2.219   | 2.147   | 2.071   | 1.989   | 1.946   | 1.901   | 1.853   | 1.802   | 1.748   | 1.690    |
| 27                                | 4.210   | 3.354   | 2.960   | 2.727   | 2.571   | 2.459   | 2.373   | 2.305   | 2.250   | 2.204   | 2.132   | 2.055   | 1.973   | 1.929   | 1.884   | 1.836   | 1.785   | 1.730   | 1.671    |
| 28                                | 4.196   | 3.340   | 2.946   | 2.714   | 2.558   | 2.445   | 2.359   | 2.291   | 2.236   | 2.190   | 2.117   | 2.041   | 1.958   | 1.914   | 1.868   | 1.820   | 1.768   | 1.713   | 1.654    |
| 29                                | 4.183   | 3.327   | 2.934   | 2.701   | 2.545   | 2.432   | 2.346   | 2.278   | 2.222   | 2.176   | 2.104   | 2.027   | 1.944   | 1.900   | 1.854   | 1.805   | 1.753   | 1.698   | 1.637    |
| 30                                | 4.170   | 3.315   | 2.922   | 2.689   | 2.533   | 2.420   | 2.334   | 2.266   | 2.210   | 2.164   | 2.092   | 2.014   | 1.931   | 1.887   | 1.840   | 1.791   | 1.739   | 1.683   | 1.622    |
| 40                                | 4.084   | 3.231   | 2.838   | 2.606   | 2.449   | 2.335   | 2.249   | 2.180   | 2.124   | 2.077   | 2.003   | 1.924   | 1.838   | 1.792   | 1.744   | 1.692   | 1.637   | 1.576   | 1.508    |
| 60                                | 4.001   | 3.150   | 2.758   | 2.525   | 2.368   | 2.254   | 2.166   | 2.097   | 2.040   | 1.992   | 1.917   | 1.836   | 1.748   | 1.700   | 1.649   | 1.594   | 1.534   | 1.467   | 1.389    |
| 120                               | 3.920   | 3.071   | 2.680   | 2.447   | 2.289   | 2.175   | 2.086   | 2.016   | 1.958   | 1.910   | 1.833   | 1.750   | 1.658   | 1.608   | 1.554   | 1.495   | 1.429   | 1.351   | 1.253    |
| $\infty$                          | 3.841   | 2.995   | 2.604   | 2.371   | 2.214   | 2.098   | 2.009   | 1.938   | 1.879   | 1.830   | 1.752   | 1.666   | 1.570   | 1.517   | 1.459   | 1.394   | 1.318   | 1.221   | 1.000    |

Degree of Freedom for Denominator



F Table for  $\alpha = 0.01$ 

Degree of Freedom for Numerator

| $df_1 \rightarrow$<br>$df_2 \downarrow$ | 1        | 2        | 3        | 4        | 5        | 6        | 7        | 8        | 9        | 10       | 12       | 15       | 20       | 24       | 30       | 40       | 60       | 120      | $\infty$ |
|-----------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| 1                                       | 4052.181 | 4999.500 | 5403.352 | 5624.583 | 5763.650 | 5858.986 | 5928.356 | 5981.070 | 6022.473 | 6055.847 | 6106.321 | 6157.285 | 6208.730 | 6234.631 | 6260.649 | 6288.782 | 6313.020 | 6339.391 | 6365.864 |
| 2                                       | 98.503   | 99.000   | 99.166   | 99.249   | 99.299   | 99.333   | 99.356   | 99.374   | 99.388   | 99.399   | 99.416   | 99.433   | 99.449   | 99.458   | 99.466   | 99.474   | 99.482   | 99.491   | 99.499   |
| 3                                       | 34.116   | 30.817   | 29.457   | 28.710   | 28.237   | 27.911   | 27.672   | 27.489   | 27.345   | 27.229   | 27.052   | 26.872   | 26.690   | 26.598   | 26.505   | 26.411   | 26.316   | 26.221   | 26.125   |
| 4                                       | 21.198   | 18.000   | 16.694   | 15.977   | 15.522   | 15.207   | 14.976   | 14.799   | 14.659   | 14.546   | 14.374   | 14.198   | 14.020   | 13.929   | 13.838   | 13.745   | 13.652   | 13.558   | 13.463   |
| 5                                       | 16.258   | 13.274   | 12.060   | 11.392   | 10.967   | 10.672   | 10.456   | 10.289   | 10.158   | 10.051   | 9.889    | 9.722    | 9.553    | 9.466    | 9.379    | 9.291    | 9.202    | 9.112    | 9.020    |
| 6                                       | 13.745   | 10.925   | 9.780    | 9.148    | 8.746    | 8.465    | 8.260    | 8.102    | 7.976    | 7.874    | 7.718    | 7.559    | 7.396    | 7.313    | 7.229    | 7.143    | 7.057    | 6.969    | 6.880    |
| 7                                       | 12.246   | 9.547    | 8.451    | 7.847    | 7.460    | 7.191    | 6.993    | 6.840    | 6.719    | 6.620    | 6.469    | 6.314    | 6.155    | 6.074    | 5.992    | 5.908    | 5.824    | 5.737    | 5.650    |
| 8                                       | 11.259   | 8.649    | 7.591    | 7.006    | 6.632    | 6.371    | 6.178    | 6.029    | 5.911    | 5.814    | 5.667    | 5.515    | 5.359    | 5.279    | 5.198    | 5.116    | 5.032    | 4.946    | 4.859    |
| 9                                       | 10.561   | 8.022    | 6.992    | 6.422    | 6.057    | 5.802    | 5.613    | 5.467    | 5.351    | 5.257    | 5.111    | 4.962    | 4.808    | 4.729    | 4.649    | 4.567    | 4.483    | 4.398    | 4.311    |
| 10                                      | 10.044   | 7.559    | 6.552    | 5.994    | 5.636    | 5.386    | 5.200    | 5.057    | 4.942    | 4.849    | 4.706    | 4.558    | 4.405    | 4.327    | 4.247    | 4.165    | 4.082    | 3.996    | 3.909    |
| 11                                      | 9.646    | 7.206    | 6.217    | 5.668    | 5.316    | 5.069    | 4.886    | 4.744    | 4.632    | 4.539    | 4.397    | 4.251    | 4.099    | 4.021    | 3.941    | 3.860    | 3.776    | 3.690    | 3.602    |
| 12                                      | 9.330    | 6.927    | 5.953    | 5.412    | 5.064    | 4.821    | 4.640    | 4.499    | 4.388    | 4.296    | 4.155    | 4.010    | 3.858    | 3.780    | 3.701    | 3.619    | 3.535    | 3.449    | 3.361    |
| 13                                      | 9.074    | 6.701    | 5.739    | 5.205    | 4.862    | 4.620    | 4.441    | 4.302    | 4.191    | 4.100    | 3.960    | 3.815    | 3.665    | 3.587    | 3.507    | 3.425    | 3.341    | 3.255    | 3.165    |
| 14                                      | 8.862    | 6.515    | 5.564    | 5.035    | 4.695    | 4.456    | 4.278    | 4.140    | 4.030    | 3.939    | 3.800    | 3.656    | 3.505    | 3.427    | 3.348    | 3.266    | 3.181    | 3.094    | 3.004    |
| 15                                      | 8.683    | 6.359    | 5.417    | 4.893    | 4.556    | 4.318    | 4.142    | 4.004    | 3.895    | 3.805    | 3.666    | 3.522    | 3.372    | 3.294    | 3.214    | 3.132    | 3.047    | 2.959    | 2.868    |
| 16                                      | 8.531    | 6.226    | 5.282    | 4.773    | 4.437    | 4.202    | 4.026    | 3.890    | 3.780    | 3.691    | 3.553    | 3.409    | 3.259    | 3.181    | 3.101    | 3.018    | 2.933    | 2.845    | 2.753    |
| 17                                      | 8.400    | 6.112    | 5.185    | 4.689    | 4.356    | 4.122    | 3.947    | 3.811    | 3.702    | 3.613    | 3.475    | 3.332    | 3.182    | 3.104    | 3.023    | 2.940    | 2.855    | 2.766    | 2.673    |
| 18                                      | 8.285    | 6.013    | 5.092    | 4.599    | 4.268    | 4.034    | 3.859    | 3.723    | 3.614    | 3.525    | 3.387    | 3.244    | 3.094    | 3.016    | 2.935    | 2.852    | 2.767    | 2.678    | 2.585    |
| 19                                      | 8.185    | 5.926    | 5.010    | 4.500    | 4.171    | 3.937    | 3.762    | 3.626    | 3.517    | 3.428    | 3.289    | 3.146    | 3.000    | 2.922    | 2.841    | 2.758    | 2.673    | 2.584    | 2.491    |
| 20                                      | 8.095    | 5.849    | 4.938    | 4.431    | 4.103    | 3.869    | 3.694    | 3.558    | 3.449    | 3.360    | 3.221    | 3.078    | 2.932    | 2.854    | 2.773    | 2.689    | 2.604    | 2.515    | 2.422    |
| 21                                      | 8.017    | 5.780    | 4.874    | 4.369    | 4.042    | 3.808    | 3.633    | 3.497    | 3.388    | 3.299    | 3.160    | 3.017    | 2.871    | 2.793    | 2.712    | 2.628    | 2.543    | 2.454    | 2.361    |
| 22                                      | 7.945    | 5.719    | 4.817    | 4.313    | 3.988    | 3.754    | 3.579    | 3.443    | 3.334    | 3.245    | 3.106    | 2.963    | 2.817    | 2.739    | 2.658    | 2.574    | 2.489    | 2.400    | 2.307    |
| 23                                      | 7.881    | 5.664    | 4.765    | 4.264    | 3.939    | 3.705    | 3.530    | 3.394    | 3.285    | 3.196    | 3.057    | 2.914    | 2.768    | 2.690    | 2.609    | 2.525    | 2.440    | 2.351    | 2.258    |
| 24                                      | 7.823    | 5.614    | 4.718    | 4.218    | 3.895    | 3.661    | 3.486    | 3.350    | 3.241    | 3.152    | 3.013    | 2.870    | 2.724    | 2.646    | 2.565    | 2.481    | 2.396    | 2.307    | 2.214    |
| 25                                      | 7.770    | 5.568    | 4.675    | 4.177    | 3.855    | 3.621    | 3.446    | 3.310    | 3.201    | 3.112    | 2.973    | 2.830    | 2.684    | 2.606    | 2.525    | 2.441    | 2.356    | 2.267    | 2.174    |
| 26                                      | 7.721    | 5.526    | 4.637    | 4.140    | 3.818    | 3.584    | 3.409    | 3.273    | 3.164    | 3.075    | 2.936    | 2.793    | 2.647    | 2.569    | 2.488    | 2.404    | 2.319    | 2.230    | 2.137    |
| 27                                      | 7.677    | 5.488    | 4.601    | 4.106    | 3.785    | 3.551    | 3.376    | 3.240    | 3.131    | 3.042    | 2.903    | 2.760    | 2.614    | 2.536    | 2.455    | 2.371    | 2.286    | 2.197    | 2.104    |
| 28                                      | 7.636    | 5.453    | 4.568    | 4.074    | 3.754    | 3.520    | 3.345    | 3.209    | 3.100    | 3.011    | 2.872    | 2.729    | 2.583    | 2.505    | 2.424    | 2.340    | 2.255    | 2.166    | 2.073    |
| 29                                      | 7.598    | 5.420    | 4.538    | 4.045    | 3.725    | 3.491    | 3.316    | 3.180    | 3.071    | 2.982    | 2.843    | 2.700    | 2.554    | 2.476    | 2.395    | 2.311    | 2.226    | 2.137    | 2.044    |
| 30                                      | 7.562    | 5.390    | 4.510    | 4.018    | 3.699    | 3.465    | 3.290    | 3.154    | 3.045    | 2.956    | 2.817    | 2.674    | 2.528    | 2.450    | 2.369    | 2.285    | 2.200    | 2.111    | 2.018    |
| 40                                      | 7.314    | 5.179    | 4.313    | 3.828    | 3.514    | 3.280    | 3.105    | 2.969    | 2.860    | 2.771    | 2.632    | 2.489    | 2.343    | 2.265    | 2.203    | 2.119    | 2.034    | 1.945    | 1.852    |
| 60                                      | 7.077    | 4.977    | 4.126    | 3.649    | 3.339    | 3.105    | 2.930    | 2.800    | 2.711    | 2.622    | 2.483    | 2.340    | 2.194    | 2.116    | 2.054    | 1.970    | 1.885    | 1.796    | 1.703    |
| 120                                     | 6.851    | 4.787    | 3.949    | 3.480    | 3.174    | 2.940    | 2.765    | 2.635    | 2.546    | 2.457    | 2.318    | 2.175    | 2.029    | 1.951    | 1.889    | 1.805    | 1.720    | 1.631    | 1.538    |
| $\infty$                                | 6.635    | 4.605    | 3.782    | 3.319    | 3.017    | 2.782    | 2.607    | 2.511    | 2.407    | 2.321    | 2.185    | 2.039    | 1.878    | 1.791    | 1.696    | 1.592    | 1.473    | 1.325    | 1.000    |

Degree of Freedom for Denominator



## Critical Values of T in the Wilcoxon Matched-Pairs Signed-Ranks Test.

| n  | Level of significance for one-tailed test |      |     |      |
|----|-------------------------------------------|------|-----|------|
|    | 0.05                                      | .025 | .01 | .005 |
|    | Level of significance for two-tailed test |      |     |      |
|    | 0.10                                      | .05  | .02 | .01  |
| 5  | 1                                         | -    | -   | -    |
| 6  | 2                                         | 1    | -   | -    |
| 7  | 4                                         | 2    | 0   | -    |
| 8  | 6                                         | 4    | 2   | 0    |
| 9  | 8                                         | 6    | 3   | 2    |
| 10 | 11                                        | 8    | 5   | 3    |
| 11 | 14                                        | 11   | 7   | 6    |
| 12 | 17                                        | 14   | 10  | 7    |
| 13 | 21                                        | 17   | 13  | 10   |
| 14 | 26                                        | 21   | 16  | 13   |
| 15 | 30                                        | 25   | 20  | 16   |
| 16 | 36                                        | 30   | 24  | 20   |
| 17 | 41                                        | 35   | 28  | 23   |
| 18 | 47                                        | 40   | 33  | 28   |
| 19 | 54                                        | 46   | 38  | 32   |
| 20 | 60                                        | 52   | 43  | 38   |
| 21 | 68                                        | 59   | 49  | 43   |
| 22 | 75                                        | 66   | 56  | 49   |
| 23 | 83                                        | 73   | 62  | 55   |
| 24 | 92                                        | 81   | 69  | 61   |
| 25 | 101                                       | 90   | 77  | 68   |



Critical values of the Kolmogorov-Smirnov One Sample Test statistics  $D_n$ .

This table gives the values of  $D_{n,\alpha}^+$  and  $D_{n,\alpha}^-$  for which  $\alpha \geq P\{D_n^+ > D_{n,\alpha}^+\}$  and  $\alpha \geq P\{D_n^- > D_{n,\alpha}^-\}$  for some selected values of  $n$  and  $\alpha$ .

| One-Sided Test:        |                         |                         |                         |                         |                         |
|------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| $\alpha =$             | .10                     | .05                     | .025                    | .01                     | .005                    |
| Two-Sided Test:        |                         |                         |                         |                         |                         |
| $\alpha =$             | .20                     | .10                     | .05                     | .02                     | .01                     |
| $n = 1$                | .900                    | .950                    | .975                    | .990                    | .995                    |
| 2                      | .684                    | .776                    | .842                    | .900                    | .929                    |
| 3                      | .565                    | .636                    | .708                    | .785                    | .829                    |
| 4                      | .493                    | .565                    | .624                    | .689                    | .734                    |
| 5                      | .447                    | .509                    | .563                    | .627                    | .669                    |
| 6                      | .410                    | .468                    | .519                    | .577                    | .617                    |
| 7                      | .381                    | .436                    | .483                    | .538                    | .576                    |
| 8                      | .358                    | .410                    | .454                    | .507                    | .542                    |
| 9                      | .339                    | .387                    | .430                    | .480                    | .513                    |
| 10                     | .323                    | .369                    | .409                    | .457                    | .489                    |
| 11                     | .308                    | .352                    | .391                    | .437                    | .468                    |
| 12                     | .296                    | .338                    | .375                    | .419                    | .449                    |
| 13                     | .285                    | .325                    | .361                    | .404                    | .432                    |
| 14                     | .275                    | .314                    | .349                    | .390                    | .418                    |
| 15                     | .266                    | .304                    | .338                    | .377                    | .404                    |
| 16                     | .258                    | .295                    | .327                    | .366                    | .392                    |
| 17                     | .250                    | .286                    | .318                    | .355                    | .381                    |
| 18                     | .244                    | .279                    | .309                    | .346                    | .371                    |
| 19                     | .237                    | .271                    | .301                    | .337                    | .361                    |
| 20                     | .232                    | .265                    | .294                    | .329                    | .352                    |
| 21                     | .226                    | .259                    | .287                    | .321                    | .344                    |
| 22                     | .221                    | .253                    | .281                    | .314                    | .337                    |
| 23                     | .216                    | .247                    | .275                    | .307                    | .330                    |
| 24                     | .212                    | .242                    | .269                    | .301                    | .323                    |
| 25                     | .208                    | .238                    | .264                    | .295                    | .317                    |
| 26                     | .204                    | .233                    | .259                    | .290                    | .311                    |
| 27                     | .200                    | .229                    | .254                    | .284                    | .305                    |
| 28                     | .197                    | .225                    | .250                    | .279                    | .300                    |
| 29                     | .193                    | .221                    | .246                    | .275                    | .295                    |
| 30                     | .190                    | .218                    | .242                    | .270                    | .290                    |
| 31                     | .187                    | .214                    | .238                    | .266                    | .285                    |
| 32                     | .184                    | .211                    | .234                    | .262                    | .281                    |
| 33                     | .182                    | .208                    | .231                    | .258                    | .277                    |
| 34                     | .179                    | .205                    | .227                    | .254                    | .273                    |
| 35                     | .177                    | .202                    | .224                    | .251                    | .269                    |
| 36                     | .174                    | .199                    | .221                    | .247                    | .265                    |
| 37                     | .172                    | .196                    | .218                    | .244                    | .262                    |
| 38                     | .170                    | .194                    | .215                    | .241                    | .258                    |
| 39                     | .168                    | .191                    | .213                    | .238                    | .255                    |
| 40                     | .165                    | .189                    | .210                    | .235                    | .252                    |
| appro.<br>for $n > 40$ | $\frac{1.07}{\sqrt{n}}$ | $\frac{1.22}{\sqrt{n}}$ | $\frac{1.36}{\sqrt{n}}$ | $\frac{1.52}{\sqrt{n}}$ | $\frac{1.63}{\sqrt{n}}$ |



Critical values of the Kolmogorov-Smirnov Test statistics for two samples of equal size

This table gives the values of  $D_{n,n,\alpha}^*$  and  $D_{n,n,\alpha}$  for which  $\alpha \geq P(D_{n,n}^* > D_{n,n,\alpha}^*)$  and  $\alpha \geq P(D_{n,n} > D_{n,n,\alpha})$  for some selected values of  $n$  and  $\alpha$ .

| One-Sided Test:        |                         |                         |                         |                         |                         |
|------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| $\alpha =$             | .10                     | .05                     | .025                    | .01                     | .005                    |
| Two-Sided Test:        |                         |                         |                         |                         |                         |
| $\alpha =$             | .20                     | .10                     | .05                     | .02                     | .01                     |
| $n = 3$                | 2/3                     | 2/3                     |                         |                         |                         |
| 4                      | 3/4                     | 3/4                     | 3/4                     |                         |                         |
| 5                      | 3/5                     | 3/5                     | 4/5                     | 4/5                     | 4/5                     |
| 6                      | 3/6                     | 4/6                     | 4/6                     | 5/6                     | 5/6                     |
| 7                      | 4/7                     | 4/7                     | 5/7                     | 5/7                     | 5/7                     |
| 8                      | 4/8                     | 4/8                     | 5/8                     | 5/8                     | 6/8                     |
| 9                      | 4/9                     | 5/9                     | 5/9                     | 6/9                     | 6/9                     |
| 10                     | 4/10                    | 5/10                    | 6/10                    | 6/10                    | 7/10                    |
| 11                     | 5/11                    | 5/11                    | 6/11                    | 7/11                    | 7/11                    |
| 12                     | 5/12                    | 5/12                    | 6/12                    | 7/12                    | 7/12                    |
| 13                     | 5/13                    | 6/13                    | 6/13                    | 7/13                    | 8/13                    |
| 14                     | 5/14                    | 6/14                    | 7/14                    | 7/14                    | 8/14                    |
| 15                     | 5/15                    | 6/15                    | 7/15                    | 8/15                    | 8/15                    |
| 16                     | 6/16                    | 6/16                    | 7/16                    | 8/16                    | 9/16                    |
| 17                     | 6/17                    | 7/17                    | 7/17                    | 8/17                    | 9/17                    |
| 18                     | 6/18                    | 7/18                    | 8/18                    | 9/18                    | 9/18                    |
| 19                     | 6/19                    | 7/19                    | 8/19                    | 9/19                    | 9/19                    |
| 20                     | 6/20                    | 7/20                    | 8/20                    | 9/20                    | 10/20                   |
| 21                     | 6/21                    | 7/21                    | 8/21                    | 9/21                    | 10/21                   |
| 22                     | 7/22                    | 8/22                    | 8/22                    | 10/22                   | 10/22                   |
| 23                     | 7/23                    | 8/23                    | 9/23                    | 10/23                   | 10/23                   |
| 24                     | 7/24                    | 8/24                    | 9/24                    | 10/24                   | 11/24                   |
| 25                     | 7/25                    | 8/25                    | 9/25                    | 10/25                   | 11/25                   |
| 26                     | 7/26                    | 8/26                    | 9/26                    | 10/26                   | 11/26                   |
| 27                     | 7/26                    | 8/27                    | 9/27                    | 10/26                   | 11/26                   |
| 28                     | 8/28                    | 9/28                    | 10/28                   | 11/27                   | 11/27                   |
| 29                     | 8/29                    | 9/29                    | 10/29                   | 11/28                   | 12/28                   |
| 30                     | 8/30                    | 9/30                    | 10/30                   | 11/29                   | 12/29                   |
| 31                     | 8/31                    | 9/31                    | 10/31                   | 11/30                   | 12/30                   |
| 32                     | 8/32                    | 9/32                    | 10/32                   | 11/31                   | 12/31                   |
| 34                     | 8/34                    | 10/34                   | 11/34                   | 12/32                   | 12/32                   |
| 36                     | 9/36                    | 10/36                   | 11/36                   | 12/34                   | 13/34                   |
| 38                     | 9/38                    | 10/38                   | 11/38                   | 12/36                   | 13/36                   |
| 40                     | 9/40                    | 10/40                   | 12/40                   | 13/38                   | 14/38                   |
| appro.<br>for $n > 40$ | $\frac{1.52}{\sqrt{n}}$ | $\frac{1.73}{\sqrt{n}}$ | $\frac{1.92}{\sqrt{n}}$ | $\frac{2.15}{\sqrt{n}}$ | $\frac{2.30}{\sqrt{n}}$ |



Critical values of the Kolmogorov-Smirnov Test Statistic for Two Samples of Unequal Size.

This table gives the values of  $D_{n_1, n_2, \alpha}^*$  and  $D_{n_1, n_2, \alpha}$  for which  $\alpha \geq P(D_{n_1, n_2}^* > D_{n_1, n_2, \alpha}^*)$  and  $\alpha \geq P(D_{n_1, n_2} > D_{n_1, n_2, \alpha})$  for some selected values of  $n_1$  = smaller sample size  $n_2$  = larger sample size, and  $\alpha$ .

| One-Sided Test: |           |       |       |       |       |       |
|-----------------|-----------|-------|-------|-------|-------|-------|
| $\alpha =$      |           | .10   | .05   | .025  | .01   | .005  |
| Two-Sided Test: |           |       |       |       |       |       |
| $\alpha =$      |           | .20   | .10   | .05   | .02   | .01   |
| $n_1 = 1$       | $n_2 = 9$ | 17/18 |       |       |       |       |
|                 | 10        | 9/10  |       |       |       |       |
| $n_1 = 2$       | $n_2 = 3$ | 5/6   |       |       |       |       |
|                 | 4         | 3/4   |       |       |       |       |
|                 | 5         | 4/5   | 4/5   |       |       |       |
|                 | 6         | 5/6   | 5/6   |       |       |       |
|                 | 7         | 5/7   | 6/7   |       |       |       |
|                 | 8         | 3/4   | 7/8   | 7/8   |       |       |
|                 | 9         | 7/9   | 8/9   | 8/9   |       |       |
|                 | 10        | 7/10  | 4/5   | 9/10  |       |       |
| $n_1 = 3$       | $n_2 = 4$ | 3/4   | 3/4   |       |       |       |
|                 | 5         | 2/3   | 4/5   | 4/5   |       |       |
|                 | 6         | 2/3   | 2/3   | 5/6   |       |       |
|                 | 7         | 2/3   | 5/7   | 6/7   | 6/7   |       |
|                 | 8         | 5/8   | 3/4   | 3/4   | 7/8   |       |
|                 | 9         | 2/3   | 2/3   | 7/9   | 8/9   | 8/9   |
|                 | 10        | 3/5   | 7/10  | 4/5   | 9/10  | 9/10  |
|                 | 12        | 7/12  | 2/3   | 3/4   | 5/6   | 11/12 |
| $n_1 = 4$       | $n_2 = 5$ | 3/5   | 3/4   | 4/5   | 4/5   |       |
|                 | 6         | 7/12  | 2/3   | 3/4   | 5/6   | 5/6   |
|                 | 7         | 17/28 | 5/7   | 3/4   | 6/7   | 6/7   |
|                 | 8         | 5/8   | 5/8   | 3/4   | 7/8   | 7/8   |
|                 | 9         | 5/9   | 2/3   | 3/4   | 7/9   | 8/9   |
|                 | 10        | 11/20 | 13/20 | 7/10  | 4/5   | 4/5   |
|                 | 12        | 7/12  | 2/3   | 2/3   | 3/4   | 5/6   |
|                 | 16        | 9/16  | 5/8   | 11/16 | 3/4   | 13/16 |
| $n_1 = 5$       | $n_2 = 6$ | 3/5   | 2/3   | 2/3   | 5/6   | 5/6   |
|                 | 7         | 4/7   | 23/35 | 5/7   | 29/35 | 6/7   |
|                 | 8         | 11/20 | 5/8   | 27/40 | 4/5   | 4/5   |
|                 | 9         | 5/9   | 3/5   | 31/45 | 7/9   | 4/5   |
|                 | 10        | 1/2   | 3/5   | 7/10  | 7/10  | 4/5   |
|                 | 15        | 8/15  | 3/5   | 2/3   | 11/15 | 11/15 |
|                 | 20        | 1/2   | 11/20 | 3/5   | 7/10  | 3/4   |



|                      |            |                                        |                                        |                                        |                                        |                                        |
|----------------------|------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|
| $n_1 = 6$            | $n_2 = 7$  | 23/42                                  | 4/7                                    | 29/42                                  | 5/7                                    | 5/6                                    |
|                      | 8          | 1/2                                    | 7/12                                   | 2/3                                    | 3/4                                    | 3/4                                    |
|                      | 9          | 1/2                                    | 5/9                                    | 2/3                                    | 13/18                                  | 7/9                                    |
|                      | 10         | 1/2                                    | 17/30                                  | 19/30                                  | 7/10                                   | 11/15                                  |
|                      | 12         | 1/2                                    | 7/12                                   | 7/12                                   | 2/3                                    | 3/4                                    |
|                      | 18         | 4/9                                    | 5/9                                    | 11/18                                  | 2/3                                    | 13/18                                  |
|                      | 24         | 11/24                                  | 1/2                                    | 7/12                                   | 5/8                                    | 2/3                                    |
| $n_1 = 7$            | $n_2 = 8$  | 27/56                                  | 33/56                                  | 5/8                                    | 41/56                                  | 3/4                                    |
|                      | 9          | 31/63                                  | 5/9                                    | 40/63                                  | 5/7                                    | 47/63                                  |
|                      | 10         | 33/70                                  | 39/70                                  | 43/70                                  | 7/10                                   | 5/7                                    |
|                      | 14         | 3/7                                    | 1/2                                    | 4/7                                    | 9/14                                   | 5/7                                    |
|                      | 28         | 3/7                                    | 13/28                                  | 15/28                                  | 17/28                                  | 9/14                                   |
| $n_1 = 8$            | $n_2 = 9$  | 4/9                                    | 13/24                                  | 5/8                                    | 2/3                                    | 3/4                                    |
|                      | 10         | 19/40                                  | 21/40                                  | 23/40                                  | 27/40                                  | 7/10                                   |
|                      | 12         | 11/24                                  | 1/2                                    | 7/12                                   | 5/8                                    | 2/3                                    |
|                      | 16         | 7/16                                   | 1/2                                    | 9/16                                   | 5/8                                    | 5/8                                    |
|                      | 32         | 13/32                                  | 7/16                                   | 1/2                                    | 9/16                                   | 19/32                                  |
| $n_1 = 9$            | $n_2 = 10$ | 7/15                                   | 1/2                                    | 26/45                                  | 2/3                                    | 31/45                                  |
|                      | 12         | 4/9                                    | 1/2                                    | 5/9                                    | 11/18                                  | 2/3                                    |
|                      | 15         | 19/45                                  | 22/45                                  | 8/15                                   | 3/5                                    | 29/45                                  |
|                      | 18         | 7/18                                   | 4/9                                    | 1/2                                    | 5/9                                    | 11/18                                  |
|                      | 36         | 13/36                                  | 5/12                                   | 17/36                                  | 19/36                                  | 5/9                                    |
| $n_1 = 10$           | $n_2 = 15$ | 2/5                                    | 7/15                                   | 1/2                                    | 17/30                                  | 19/30                                  |
|                      | 20         | 2/5                                    | 9/20                                   | 1/2                                    | 11/20                                  | 3/5                                    |
|                      | 40         | 7/20                                   | 2/5                                    | 9/20                                   | 1/2                                    |                                        |
| $n_1 = 12$           | $n_2 = 15$ | 23/60                                  | 9/20                                   | 1/2                                    | 11/20                                  | 7/12                                   |
|                      | 16         | 3/8                                    | 7/16                                   | 23/48                                  | 13/24                                  | 7/12                                   |
|                      | 18         | 11/36                                  | 5/12                                   | 17/36                                  | 19/36                                  | 5/9                                    |
|                      | 20         | 13/30                                  | 5/12                                   | 7/15                                   | 31/60                                  | 17/30                                  |
| $n_1 = 15$           | $n_2 = 20$ | 7/20                                   | 2/5                                    | 13/30                                  | 29/60                                  | 31/60                                  |
| $n_1 = 16$           | $n_2 = 20$ | 27/80                                  | 31/80                                  | 17/40                                  | 19/40                                  | 41/80                                  |
| Large sample approx. |            | $1.07\sqrt{\frac{n_1 + n_2}{n_1 n_2}}$ | $1.22\sqrt{\frac{n_1 + n_2}{n_1 n_2}}$ | $1.36\sqrt{\frac{n_1 + n_2}{n_1 n_2}}$ | $1.52\sqrt{\frac{n_1 + n_2}{n_1 n_2}}$ | $1.63\sqrt{\frac{n_1 + n_2}{n_1 n_2}}$ |



Probabilities associated with values as large as observed values of Friedman Statistic  $F_r$   
 i.e.  $p_0 = P(F_r > F_r^*)$  where  $F_r^*$  is the calculated value of  $F_r$ .

Table for  $k = 3$ 

| n = 2 |       | n = 3 |       | n = 4 |       | n = 5 |        |
|-------|-------|-------|-------|-------|-------|-------|--------|
| $F_r$ | p     | $F_r$ | p     | $F_r$ | p     | $F_r$ | p      |
| 0     | 1.000 | .000  | 1.000 | .0    | 1.000 | .0    | 1.000  |
| 1     | .833  | .667  | .994  | .5    | .931  | .4    | .954   |
| 3     | .500  | 2.000 | .528  | 1.5   | .653  | 1.2   | .691   |
| 4     | .167  | 2.667 | .361  | 2.0   | .431  | 1.6   | .522   |
|       |       | 4.667 | .194  | 3.5   | .273  | 2.8   | .367   |
|       |       | 6.000 | .028  | 4.5   | .125  | 3.6   | .182   |
|       |       |       |       | 6.0   | .069  | 4.8   | .124   |
|       |       |       |       | 6.5   | .042  | 5.2   | .093   |
|       |       |       |       | 8.0   | .0046 | 6.4   | .039   |
|       |       |       |       |       |       | 7.6   | .024   |
|       |       |       |       |       |       | 8.4   | .0085  |
|       |       |       |       |       |       | 10.0  | .00077 |

| n = 6 |        | n = 7  |         | n = 8 |          | n = 9  |          |
|-------|--------|--------|---------|-------|----------|--------|----------|
| $F_r$ | p      | $F_r$  | p       | $F_r$ | p        | $F_r$  | p        |
| .00   | 1.000  | .000   | 1.000   | .00   | 1.000    | .000   | 1.000    |
| .33   | .956   | .286   | .964    | .25   | .967     | .222   | .971     |
| 1.00  | .740   | .857   | .768    | .75   | .794     | .667   | .814     |
| 1.33  | .570   | 1.143  | .620    | 1.00  | .654     | .889   | .865     |
| 2.33  | .430   | 2.000  | .486    | 1.75  | .531     | 1.556  | .569     |
| 3.00  | .252   | 2.571  | .305    | 2.25  | .355     | 2.000  | .398     |
| 4.20  | .184   | 3.429  | .237    | 3.00  | .285     | 2.667  | .328     |
| 4.33  | .142   | 3.714  | .192    | 3.25  | .236     | 2.889  | .278     |
| 5.33  | .072   | 4.571  | .112    | 4.00  | .149     | 3.556  | .187     |
| 6.33  | .052   | 5.429  | .085    | 4.75  | .120     | 4.222  | .154     |
| 7.00  | .029   | 6.000  | .052    | 5.25  | .079     | 4.667  | .107     |
| 8.33  | .012   | 7.143  | .027    | 6.25  | .047     | 5.556  | .069     |
| 9.00  | .0081  | 7.714  | .021    | 6.75  | .038     | 6.000  | .057     |
| 9.33  | .0055  | 8.000  | .016    | 7.00  | .030     | 6.222  | .048     |
| 10.33 | .0017  | 8.857  | .0084   | 7.75  | .018     | 6.889  | .031     |
| 12.00 | .00013 | 10.286 | .0036   | 9.00  | .0099    | 8.000  | .019     |
|       |        | 10.571 | .0027   | 9.25  | .0080    | 8.222  | .016     |
|       |        | 11.143 | .0012   | 9.75  | .0048    | 8.667  | .010     |
|       |        | 12.286 | .00032  | 10.75 | .0024    | 9.556  | .0060    |
|       |        | 14.000 | .000021 | 12.00 | .0011    | 10.667 | .0035    |
|       |        |        |         | 12.25 | .00086   | 10.889 | .0029    |
|       |        |        |         | 13.00 | .00026   | 11.556 | .0013    |
|       |        |        |         | 14.25 | .000061  | 12.667 | .00066   |
|       |        |        |         | 16.00 | .0000036 | 13.556 | .00035   |
|       |        |        |         |       |          | 14.000 | .00020   |
|       |        |        |         |       |          | 14.222 | .000097  |
|       |        |        |         |       |          | 14.889 | .000054  |
|       |        |        |         |       |          | 16.222 | .000011  |
|       |        |        |         |       |          | 18.000 | .0000006 |



Table for  $k = 4$ 

| n=2   |       | n=3   |       | n=4   |       | n=5   |         |
|-------|-------|-------|-------|-------|-------|-------|---------|
| $F_r$ | p     | $F_r$ | p     | $F_r$ | p     | $F_r$ | p       |
| .0    | 1.000 | .2    | 1.000 | .0    | 1.000 | 5.7   | .141    |
| .6    | .958  | .6    | .958  | .3    | .992  | 6.0   | .105    |
| 1.2   | .834  | 1.0   | .910  | .6    | .928  | 6.3   | .094    |
| 1.8   | .792  | 1.8   | .727  | .9    | .900  | 6.6   | .077    |
| 2.4   | .625  | 2.2   | .608  | 1.2   | .800  | 6.9   | .068    |
| 3.0   | .542  | 2.6   | .524  | 1.5   | .754  | 7.2   | .054    |
| 3.6   | .458  | 3.4   | .446  | 1.8   | .677  | 7.5   | .052    |
| 4.2   | .375  | 3.8   | .342  | 2.1   | .649  | 7.8   | .036    |
| 4.8   | .208  | 4.2   | .300  | 2.4   | .524  | 8.1   | .033    |
| 5.4   | .167  | 5.0   | .207  | 2.7   | .508  | 8.4   | .019    |
| 6.0   | .042  | 5.4   | .175  | 3.0   | .432  | 8.7   | .014    |
|       |       | 5.8   | .148  | 3.3   | .389  | 9.3   | .012    |
|       |       | 6.6   | .075  | 3.6   | .355  | 9.6   | .0069   |
|       |       | 7.0   | .054  | 3.9   | .324  | 9.9   | .0062   |
|       |       | 7.4   | .033  | 4.5   | .242  | 10.2  | .0027   |
|       |       | 8.2   | .017  | 4.8   | .200  | 10.8  | .0016   |
|       |       | 9.0   | .0017 | 5.1   | .190  | 11.1  | .00094  |
|       |       |       |       | 5.4   | .158  | 12.0  | .000072 |



### Critical values for total number of runs 'r' at $\alpha = 0.05$ for two tailed test.

The smaller critical value for a left-hand critical region, the larger for a right-hand critical region. For a one tailed test  $\alpha = 0.025$  and use only one of the critical values of r.

|                                |    | The larger of $n_1$ and $n_2$ |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |
|--------------------------------|----|-------------------------------|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|
|                                |    | 5                             | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| The Smaller of $n_1$ and $n_2$ | 2  |                               |   |   |   |   |    |    | 2  | 2  | 2  | 2  | 2  | 2  | 2  | 2  | 2  |
|                                | 3  |                               | 2 | 2 | 2 | 2 | 2  | 2  | 2  | 2  | 2  | 3  | 3  | 3  | 3  | 3  | 3  |
|                                | 4  | 2                             | 2 | 2 | 3 | 3 | 3  | 3  | 3  | 3  | 3  | 3  | 4  | 4  | 4  | 4  | 4  |
|                                | 5  | 2                             | 3 | 3 | 3 | 3 | 3  | 4  | 4  | 4  | 4  | 4  | 4  | 4  | 5  | 5  | 5  |
|                                | 6  |                               | 3 | 3 | 3 | 4 | 4  | 4  | 4  | 5  | 5  | 5  | 5  | 5  | 5  | 6  | 6  |
|                                | 7  |                               |   | 3 | 4 | 4 | 5  | 5  | 5  | 5  | 6  | 6  | 6  | 6  | 6  | 6  | 6  |
|                                | 8  |                               |   |   | 4 | 5 | 5  | 5  | 6  | 6  | 6  | 6  | 6  | 7  | 7  | 7  | 7  |
|                                | 9  |                               |   |   |   | 5 | 5  | 6  | 6  | 6  | 7  | 7  | 7  | 7  | 7  | 7  | 7  |
|                                | 10 |                               |   |   |   |   | 6  | 6  | 7  | 7  | 7  | 8  | 8  | 8  | 8  | 9  | 9  |
|                                | 11 |                               |   |   |   |   |    | 7  | 7  | 8  | 8  | 8  | 9  | 9  | 9  | 9  | 9  |
|                                | 12 |                               |   |   |   |   |    |    | 7  | 8  | 8  | 9  | 9  | 9  | 10 | 10 | 10 |
|                                | 13 |                               |   |   |   |   |    |    |    | 8  | 9  | 9  | 10 | 10 | 10 | 10 | 10 |
|                                | 14 |                               |   |   |   |   |    |    |    |    | 9  | 10 | 10 | 11 | 11 | 11 | 11 |
|                                | 15 |                               |   |   |   |   |    |    |    |    |    | 10 | 11 | 11 | 11 | 12 | 12 |
|                                | 16 |                               |   |   |   |   |    |    |    |    |    |    | 11 | 11 | 12 | 12 | 12 |
|                                | 17 |                               |   |   |   |   |    |    |    |    |    |    |    | 11 | 12 | 13 | 13 |
|                                | 18 |                               |   |   |   |   |    |    |    |    |    |    |    |    | 12 | 13 | 13 |
|                                | 19 |                               |   |   |   |   |    |    |    |    |    |    |    |    |    | 13 | 13 |
|                                | 20 |                               |   |   |   |   |    |    |    |    |    |    |    |    |    |    | 14 |



### Distribution function of U i.e. $P(U \leq U_0) = p_0$ , $n_1 \leq n_2$ and $3 \leq n_2 \leq 10$

The probabilities associated with the values as small as observed value of U in the Mann-Whitney test.

$n_2 = 3$

| $U_0$ | $n_1$ |     |     |
|-------|-------|-----|-----|
|       | 1     | 2   | 3   |
| 0     | .25   | .10 | .05 |
| 1     | .50   | .20 | .10 |
| 2     |       | .40 | .20 |
| 3     |       | .60 | .35 |
| 4     |       |     | .50 |

$n_2 = 4$

| $U_0$ | $n_1$ |       |       |       |
|-------|-------|-------|-------|-------|
|       | 1     | 2     | 3     | 4     |
| 0     | .2000 | .0667 | .0286 | .0143 |
| 1     | .4000 | .1333 | .0571 | .0286 |
| 2     | .6000 | .2667 | .1143 | .0571 |
| 3     |       | .4000 | .2000 | .1000 |
| 4     |       | .6000 | .3143 | .1714 |
| 5     |       |       | .4286 | .2429 |
| 6     |       |       | .5714 | .3429 |
| 7     |       |       |       | .4429 |
| 8     |       |       |       | .5571 |

$n_2 = 5$

| $U_0$ | $n_1$ |       |       |       |       |
|-------|-------|-------|-------|-------|-------|
|       | 1     | 2     | 3     | 4     | 5     |
| 0     | .1667 | .0476 | .0179 | .0079 | .0040 |
| 1     | .3333 | .0952 | .0357 | .0159 | .0079 |
| 2     | .5000 | .1905 | .0714 | .0317 | .0159 |
| 3     |       | .2857 | .1250 | .0556 | .0278 |
| 4     |       | .4286 | .1964 | .0952 | .0476 |
| 5     |       | .5714 | .2857 | .1429 | .0754 |
| 6     |       |       | .3929 | .2063 | .1111 |
| 7     |       |       | .5000 | .2778 | .1548 |
| 8     |       |       |       | .3651 | .2103 |
| 9     |       |       |       | .4524 | .2738 |
| 10    |       |       |       | .5476 | .3452 |
| 11    |       |       |       |       | .4206 |
| 12    |       |       |       |       | .5000 |



$n_2 = 6$ 

| $U_0$ | $n_1$ |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|
|       | 1     | 2     | 3     | 4     | 5     | 6     |
| 0     | .1429 | .0357 | .0119 | .0048 | .0022 | .0011 |
| 1     | .2857 | .0714 | .0238 | .0095 | .0043 | .0022 |
| 2     | .4286 | .1429 | .0476 | .0190 | .0087 | .0043 |
| 3     | .5714 | .2143 | .0833 | .0333 | .0152 | .0076 |
| 4     |       | .3214 | .1310 | .0571 | .0260 | .0130 |
| 5     |       | .4286 | .1905 | .0857 | .0411 | .0206 |
| 6     |       | .5714 | .2738 | .1286 | .0628 | .0325 |
| 7     |       |       | .3571 | .1762 | .0887 | .0465 |
| 8     |       |       | .4524 | .2381 | .1234 | .0660 |
| 9     |       |       | .5476 | .3048 | .1645 | .0898 |
| 10    |       |       |       | .3810 | .2143 | .1201 |
| 11    |       |       |       | .4571 | .2684 | .1548 |
| 12    |       |       |       | .5429 | .3312 | .1970 |
| 13    |       |       |       |       | .3961 | .2424 |
| 14    |       |       |       |       | .4654 | .2944 |
| 15    |       |       |       |       | .5346 | .3496 |
| 16    |       |       |       |       |       | .4091 |
| 17    |       |       |       |       |       | .4686 |
| 18    |       |       |       |       |       | .5314 |

 $n_2 = 7$ 

| $U_0$ | $n_1$ |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|
|       | 1     | 2     | 3     | 4     | 5     | 6     | 7     |
| 0     | .1250 | .0278 | .0083 | .0030 | .0013 | .0006 | .0003 |
| 1     | .2500 | .0556 | .0167 | .0061 | .0025 | .0012 | .0006 |
| 2     | .3750 | .1111 | .0333 | .0121 | .0051 | .0023 | .0012 |
| 3     | .5000 | .1667 | .0583 | .0212 | .0088 | .0041 | .0020 |
| 4     |       | .2500 | .0917 | .0364 | .0152 | .0070 | .0035 |
| 5     |       | .3333 | .1333 | .0545 | .0240 | .0111 | .0055 |
| 6     |       | .4444 | .1917 | .0818 | .0366 | .0175 | .0087 |
| 7     |       | .5556 | .2583 | .1152 | .0530 | .0256 | .0131 |
| 8     |       |       | .3333 | .1576 | .0745 | .0367 | .0189 |
| 9     |       |       | .4167 | .2061 | .1010 | .0507 | .0265 |
| 10    |       |       | .5000 | .2636 | .1338 | .0688 | .0364 |
| 11    |       |       |       | .3242 | .1717 | .0903 | .0487 |
| 12    |       |       |       | .3939 | .2159 | .1171 | .0641 |
| 13    |       |       |       | .4636 | .2652 | .1474 | .0825 |
| 14    |       |       |       | .5364 | .3194 | .1830 | .1043 |
| 15    |       |       |       |       | .3775 | .2226 | .1297 |
| 16    |       |       |       |       | .4381 | .2669 | .1588 |
| 17    |       |       |       |       | .5000 | .3141 | .1914 |
| 18    |       |       |       |       |       | .3654 | .2279 |
| 19    |       |       |       |       |       | .4178 | .2675 |
| 20    |       |       |       |       |       | .4726 | .3100 |
| 21    |       |       |       |       |       | .5274 | .3552 |
| 22    |       |       |       |       |       |       | .4024 |
| 23    |       |       |       |       |       |       | .4508 |
| 24    |       |       |       |       |       |       | .5000 |



$n_2 = 8$ 

| $U_0$ | $n_1$ |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|       | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     |
| 0     | .1111 | .0222 | .0061 | .0020 | .0008 | .0003 | .0002 | .0001 |
| 1.    | .2222 | .0444 | .0121 | .0040 | .0016 | .0007 | .0003 | .0002 |
| 2.    | .3333 | .0889 | .0242 | .0081 | .0031 | .0013 | .0006 | .0003 |
| 3.    | .4444 | .1333 | .0424 | .0141 | .0054 | .0023 | .0011 | .0005 |
| 4.    | .5556 | .2000 | .0667 | .0242 | .0093 | .0040 | .0019 | .0009 |
| 5.    |       | .2667 | .0970 | .0364 | .0148 | .0063 | .0030 | .0015 |
| 6.    |       | .3556 | .1394 | .0545 | .0225 | .0100 | .0047 | .0023 |
| 7.    |       | .4444 | .1879 | .0788 | .0326 | .0147 | .0070 | .0035 |
| 8.    |       | .5556 | .2485 | .1071 | .0466 | .0213 | .0103 | .0052 |
| 9.    |       |       | .3152 | .1414 | .0637 | .0296 | .0145 | .0074 |
| 10.   |       |       | .3879 | .1838 | .0855 | .0406 | .0200 | .0103 |
| 11.   |       |       | .4606 | .2303 | .1111 | .0539 | .0270 | .0141 |
| 12.   |       |       | .5394 | .2848 | .1422 | .0709 | .0361 | .0190 |
| 13.   |       |       |       | .3414 | .1772 | .0906 | .0469 | .0249 |
| 14.   |       |       |       | .4040 | .2176 | .1142 | .0603 | .0325 |
| 15.   |       |       |       | .4667 | .2618 | .1412 | .0760 | .0415 |
| 16.   |       |       |       | .5333 | .3108 | .1725 | .0946 | .0524 |
| 17.   |       |       |       |       | .3621 | .2068 | .1159 | .0652 |
| 18.   |       |       |       |       | .4165 | .2454 | .1405 | .0803 |
| 19.   |       |       |       |       | .4716 | .2864 | .1678 | .0974 |
| 20.   |       |       |       |       | .5284 | .3310 | .1984 | .1172 |
| 21.   |       |       |       |       |       | .3773 | .2317 | .1393 |
| 22.   |       |       |       |       |       | .4259 | .2679 | .1641 |
| 23.   |       |       |       |       |       | .4749 | .3063 | .1911 |
| 24.   |       |       |       |       |       | .5251 | .3472 | .2209 |
| 25.   |       |       |       |       |       |       | .3894 | .2527 |
| 26.   |       |       |       |       |       |       | .4333 | .2869 |
| 27.   |       |       |       |       |       |       | .4775 | .3227 |
| 28.   |       |       |       |       |       |       | .5225 | .3605 |
| 29.   |       |       |       |       |       |       |       | .3992 |
| 30.   |       |       |       |       |       |       |       | .4392 |
| 31.   |       |       |       |       |       |       |       | .4796 |
| 32.   |       |       |       |       |       |       |       | .5204 |

 $n_2 = 9$ 

| $U_0$ | $n_1$ |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|       | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     |
| 0     | .1000 | .0182 | .0045 | .0014 | .0005 | .0002 | .0001 | .0000 | .0000 |
| 1.    | .2000 | .0364 | .0091 | .0028 | .0010 | .0004 | .0002 | .0001 | .0000 |
| 2.    | .3000 | .0727 | .0182 | .0058 | .0020 | .0008 | .0003 | .0002 | .0001 |
| 3.    | .4000 | .1091 | .0318 | .0098 | .0035 | .0014 | .0006 | .0003 | .0001 |
| 4.    | .5000 | .1636 | .0500 | .0168 | .0060 | .0024 | .0010 | .0005 | .0002 |
| 5.    |       | .2182 | .0727 | .0252 | .0095 | .0038 | .0017 | .0008 | .0004 |
| 6.    |       | .2909 | .1045 | .0378 | .0145 | .0060 | .0026 | .0012 | .0006 |
| 7.    |       | .3636 | .1409 | .0531 | .0210 | .0088 | .0039 | .0019 | .0009 |
| 8.    |       | .4545 | .1864 | .0741 | .0300 | .0128 | .0058 | .0028 | .0014 |
| 9.    |       | .5455 | .2409 | .0993 | .0415 | .0180 | .0082 | .0039 | .0020 |



|     |  |  |       |       |       |       |       |       |       |
|-----|--|--|-------|-------|-------|-------|-------|-------|-------|
| 10. |  |  | .3000 | .1301 | .0559 | .0248 | .0115 | .0056 | .0028 |
| 11. |  |  | .3636 | .1650 | .0734 | .0332 | .0156 | .0076 | .0039 |
| 12. |  |  | .4318 | .2070 | .0949 | .0440 | .0209 | .0103 | .0053 |
| 13. |  |  | .5000 | .2517 | .1199 | .0567 | .0274 | .0137 | .0071 |
| 14. |  |  |       | .3021 | .1489 | .0723 | .0356 | .0180 | .0094 |
| 15. |  |  |       | .3552 | .1818 | .0905 | .0454 | .0232 | .0122 |
| 16. |  |  |       | .4126 | .2188 | .1119 | .0571 | .0296 | .0157 |
| 17. |  |  |       | .4699 | .2592 | .1361 | .0708 | .0372 | .0200 |
| 18. |  |  |       | .5301 | .3032 | .1638 | .0869 | .0464 | .0252 |
| 19. |  |  |       |       | .3497 | .1942 | .1052 | .0570 | .0313 |
| 20. |  |  |       |       | .3986 | .2280 | .1261 | .0694 | .0385 |
| 21. |  |  |       |       | .4491 | .2643 | .1496 | .0836 | .0470 |
| 22. |  |  |       |       | .5000 | .3035 | .1755 | .0998 | .0567 |
| 23. |  |  |       |       |       | .3445 | .2039 | .1179 | .0680 |
| 24. |  |  |       |       |       | .3878 | .2349 | .1383 | .0807 |
| 25. |  |  |       |       |       | .4320 | .2680 | .1606 | .0951 |
| 26. |  |  |       |       |       | .4773 | .3032 | .1852 | .1112 |
| 27. |  |  |       |       |       | .5227 | .3403 | .2117 | .1290 |
| 28. |  |  |       |       |       |       | .3788 | .2404 | .1487 |
| 29. |  |  |       |       |       |       | .4185 | .2707 | .1701 |
| 30. |  |  |       |       |       |       | .4591 | .3029 | .1933 |
| 31. |  |  |       |       |       |       | .5000 | .3365 | .2181 |
| 32. |  |  |       |       |       |       |       | .3715 | .2447 |
| 33. |  |  |       |       |       |       |       | .4074 | .2729 |
| 34. |  |  |       |       |       |       |       | .4442 | .3024 |
| 35. |  |  |       |       |       |       |       | .4813 | .3332 |
| 36. |  |  |       |       |       |       |       | .5187 | .3652 |
| 37. |  |  |       |       |       |       |       |       | .3981 |
| 38. |  |  |       |       |       |       |       |       | .4317 |
| 39. |  |  |       |       |       |       |       |       | .4657 |
| 40. |  |  |       |       |       |       |       |       | .5000 |

 $n_2 = 10$ 

| $U_0$ | $n_1$ |       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|       | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |
| 0     | .0909 | .0152 | .0035 | .0010 | .0003 | .0001 | .0001 | .0000 | .0000 | .0000 |
| 1.    | .1818 | .0303 | .0070 | .0020 | .0007 | .0002 | .0001 | .0000 | .0000 | .0000 |
| 2.    | .2727 | .0606 | .0140 | .0040 | .0013 | .0005 | .0002 | .0001 | .0000 | .0000 |
| 3.    | .3636 | .0909 | .0245 | .0070 | .0023 | .0009 | .0004 | .0002 | .0001 | .0000 |
| 4.    | .4545 | .1364 | .0385 | .0120 | .0040 | .0015 | .0006 | .0003 | .0001 | .0001 |
| 5.    | .5455 | .1818 | .0559 | .0180 | .0063 | .0024 | .0010 | .0004 | .0002 | .0001 |
| 6.    |       | .2424 | .0804 | .0270 | .0097 | .0037 | .0015 | .0007 | .0003 | .0002 |
| 7.    |       | .3030 | .1084 | .0380 | .0140 | .0055 | .0023 | .0010 | .0005 | .0002 |
| 8.    |       | .3788 | .1434 | .0529 | .0200 | .0080 | .0034 | .0015 | .0007 | .0004 |
| 9.    |       | .4545 | .1853 | .0709 | .0276 | .0112 | .0048 | .0022 | .0011 | .0005 |
| 10.   |       | .5455 | .2343 | .0939 | .0376 | .0158 | .0068 | .0031 | .0015 | .0008 |
| 11.   |       |       | .2867 | .1199 | .0498 | .0210 | .0093 | .0043 | .0021 | .0010 |



|     |  |       |       |       |       |       |       |       |       |
|-----|--|-------|-------|-------|-------|-------|-------|-------|-------|
| 12. |  | .3462 | .1518 | .0646 | .0280 | .0125 | .0058 | .0028 | .0014 |
| 13. |  | .4056 | .1868 | .0823 | .0363 | .0165 | .0078 | .0038 | .0019 |
| 14. |  | .4685 | .2268 | .1032 | .0467 | .0215 | .0103 | .0051 | .0026 |
| 15. |  | .5315 | .2697 | .1272 | .0589 | .0277 | .0133 | 0.066 | .0034 |
| 16. |  |       | .3177 | .1548 | .0736 | .0351 | .0171 | .0086 | .0045 |
| 17. |  |       | .3666 | .1855 | .0903 | .0439 | .0217 | .0110 | .0057 |
| 18. |  |       | .4196 | .2198 | .1099 | .0544 | .0273 | .0140 | .0073 |
| 19. |  |       | .4725 | .2567 | .1317 | .0665 | .0338 | .0175 | .0093 |
| 20. |  |       | .5275 | .2970 | .1566 | .0806 | .0416 | .0217 | .0116 |
| 21. |  |       |       | .3393 | .1838 | .0966 | .0506 | .0267 | .0144 |
| 22. |  |       |       | .3839 | .2139 | .1148 | .0610 | .0326 | .0177 |
| 23. |  |       |       | .4296 | .2461 | .1349 | .0729 | .0394 | .0216 |
| 24. |  |       |       | .4765 | .2811 | .1574 | .0864 | .0474 | .0262 |
| 25. |  |       |       | .5235 | .3177 | .1819 | .1015 | .0564 | .0315 |
| 26. |  |       |       |       | .3564 | .2087 | .1185 | .0667 | .0376 |
| 27. |  |       |       |       | .3962 | .2374 | .1371 | .0782 | .0446 |
| 28. |  |       |       |       | .4374 | .2681 | .1577 | .0912 | .0526 |
| 29. |  |       |       |       | .4789 | .3004 | .1800 | .1055 | .0615 |
| 30. |  |       |       |       | .5211 | .3345 | .2041 | .1214 | .0716 |
| 31. |  |       |       |       |       | .3698 | .2299 | .1388 | .0827 |
| 32. |  |       |       |       |       | .4063 | .2574 | .1577 | .0952 |
| 33. |  |       |       |       |       | .4434 | .2863 | .1781 | .1088 |
| 34. |  |       |       |       |       | .4811 | .3167 | .2001 | .1237 |
| 35. |  |       |       |       |       | .5189 | .3482 | .2235 | .1399 |
| 36. |  |       |       |       |       |       | .3809 | .2483 | .1575 |
| 37. |  |       |       |       |       |       | .4143 | .2745 | .1763 |
| 38. |  |       |       |       |       |       | .4484 | .3019 | .1965 |
| 39. |  |       |       |       |       |       | .4827 | .3304 | .2179 |
| 40. |  |       |       |       |       |       | .5173 | .3598 | .2406 |
| 41. |  |       |       |       |       |       |       | .3901 | .2644 |
| 42. |  |       |       |       |       |       |       | .4211 | .2894 |
| 43. |  |       |       |       |       |       |       | .4524 | .3153 |
| 44. |  |       |       |       |       |       |       | .4841 | .3421 |
| 45. |  |       |       |       |       |       |       | .5159 | .3697 |
| 46. |  |       |       |       |       |       |       |       | .3980 |
| 47. |  |       |       |       |       |       |       |       | .4267 |
| 48. |  |       |       |       |       |       |       |       | .4559 |
| 49. |  |       |       |       |       |       |       |       | .4853 |
| 50. |  |       |       |       |       |       |       |       | .5147 |



# Critical value of U in the Mann-Whitney Test

a. Critical values of U for a one-tailed test at 0.025 or for a two-tailed test at 0.05

| $n_1$<br>$n_2$ | 1 | 2 | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16  | 17  | 18  | 19  | 20 |
|----------------|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|-----|-----|-----|----|
| 1              |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |     |     |     |     |    |
| 2              |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |     |     |     |     |    |
| 3              |   |   |    |    | 0  | 1  | 1  | 2  | 2  | 3  | 3  | 4  | 4  | 5  | 5  | 6   | 6   | 7   | 7   | 8  |
| 4              |   |   | 0  | 1  | 2  | 3  | 4  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 11  | 12  | 13  | 13  |    |
| 5              |   | 0 | 1  | 2  | 3  | 5  | 6  | 7  | 8  | 9  | 11 | 12 | 13 | 14 | 15 | 17  | 18  | 19  | 20  |    |
| 6              |   | 1 | 2  | 3  | 5  | 6  | 8  | 10 | 11 | 13 | 14 | 16 | 17 | 19 | 21 | 22  | 24  | 25  | 27  |    |
| 7              |   | 1 | 3  | 5  | 6  | 8  | 10 | 12 | 14 | 16 | 18 | 20 | 22 | 24 | 26 | 28  | 30  | 32  | 34  |    |
| 8              | 0 | 2 | 4  | 6  | 8  | 10 | 13 | 15 | 17 | 19 | 22 | 24 | 26 | 29 | 31 | 34  | 36  | 38  | 41  |    |
| 9              | 0 | 2 | 4  | 7  | 10 | 12 | 15 | 17 | 20 | 23 | 26 | 28 | 31 | 34 | 37 | 39  | 42  | 45  | 48  |    |
| 10             | 0 | 3 | 5  | 8  | 11 | 14 | 17 | 20 | 23 | 26 | 29 | 33 | 36 | 39 | 42 | 45  | 48  | 52  | 55  |    |
| 11             | 0 | 3 | 6  | 9  | 13 | 16 | 19 | 23 | 26 | 30 | 33 | 37 | 40 | 44 | 47 | 51  | 55  | 58  | 62  |    |
| 12             | 1 | 4 | 7  | 11 | 14 | 18 | 22 | 26 | 29 | 33 | 37 | 41 | 45 | 49 | 53 | 57  | 61  | 65  | 69  |    |
| 13             | 1 | 4 | 8  | 12 | 16 | 20 | 24 | 28 | 33 | 37 | 41 | 45 | 50 | 54 | 59 | 63  | 67  | 72  | 76  |    |
| 14             | 1 | 5 | 9  | 13 | 17 | 22 | 26 | 31 | 36 | 40 | 45 | 50 | 55 | 59 | 64 | 67  | 74  | 78  | 83  |    |
| 15             | 1 | 5 | 10 | 14 | 19 | 24 | 29 | 34 | 39 | 44 | 49 | 54 | 59 | 64 | 70 | 75  | 80  | 85  | 90  |    |
| 16             | 1 | 6 | 11 | 15 | 21 | 26 | 31 | 37 | 42 | 47 | 53 | 59 | 64 | 70 | 75 | 81  | 86  | 92  | 98  |    |
| 17             | 2 | 6 | 11 | 17 | 22 | 28 | 34 | 39 | 45 | 51 | 57 | 63 | 67 | 75 | 81 | 87  | 93  | 99  | 105 |    |
| 18             | 2 | 7 | 12 | 18 | 24 | 30 | 36 | 42 | 48 | 55 | 61 | 67 | 74 | 80 | 86 | 93  | 99  | 106 | 112 |    |
| 19             | 2 | 7 | 13 | 19 | 25 | 32 | 38 | 45 | 52 | 58 | 65 | 72 | 78 | 85 | 92 | 99  | 106 | 113 | 119 |    |
| 20             | 2 | 8 | 13 | 20 | 27 | 34 | 41 | 48 | 55 | 62 | 69 | 76 | 83 | 90 | 98 | 103 | 112 | 119 | 127 |    |

b. Critical values of U for a one-tailed test at 0.05 or for a two-tailed test at 0.10

| $n_1$<br>$n_2$ | 1 | 2 | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15  | 16  | 17  | 18  | 19  | 20  |
|----------------|---|---|----|----|----|----|----|----|----|----|----|----|----|----|-----|-----|-----|-----|-----|-----|
| 1              |   |   |    |    |    |    |    |    |    |    |    |    |    |    |     |     |     |     | 0   | 0   |
| 2              |   |   |    |    | 0  | 0  | 0  | 1  | 1  | 1  | 1  | 2  | 2  | 2  | 3   | 3   | 3   | 4   | 4   | 4   |
| 3              |   |   | 0  | 0  | 1  | 2  | 2  | 3  | 3  | 4  | 5  | 5  | 6  | 7  | 7   | 8   | 9   | 9   | 10  | 11  |
| 4              |   |   | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12  | 14  | 15  | 16  | 17  | 18  |
| 5              |   | 0 | 1  | 2  | 4  | 5  | 6  | 8  | 9  | 11 | 12 | 13 | 15 | 16 | 18  | 19  | 20  | 22  | 23  | 25  |
| 6              |   | 0 | 2  | 3  | 5  | 7  | 8  | 10 | 12 | 14 | 16 | 17 | 19 | 21 | 23  | 25  | 26  | 28  | 30  | 32  |
| 7              |   | 0 | 2  | 4  | 6  | 8  | 11 | 13 | 15 | 17 | 19 | 21 | 24 | 26 | 28  | 30  | 33  | 35  | 37  | 39  |
| 8              |   | 1 | 3  | 5  | 8  | 10 | 13 | 15 | 18 | 20 | 23 | 26 | 28 | 31 | 33  | 36  | 39  | 41  | 44  | 47  |
| 9              |   | 1 | 3  | 6  | 9  | 12 | 15 | 18 | 21 | 24 | 27 | 30 | 33 | 36 | 39  | 42  | 45  | 48  | 51  | 54  |
| 10             |   | 1 | 4  | 7  | 11 | 14 | 17 | 20 | 24 | 27 | 31 | 34 | 37 | 41 | 44  | 48  | 51  | 55  | 58  | 62  |
| 11             |   | 1 | 5  | 8  | 12 | 16 | 19 | 23 | 27 | 31 | 34 | 38 | 42 | 46 | 50  | 54  | 57  | 61  | 65  | 69  |
| 12             |   | 2 | 5  | 9  | 13 | 17 | 21 | 26 | 30 | 34 | 38 | 42 | 47 | 51 | 55  | 60  | 64  | 68  | 72  | 77  |
| 13             |   | 2 | 6  | 10 | 15 | 19 | 24 | 28 | 33 | 37 | 42 | 47 | 51 | 56 | 61  | 65  | 70  | 75  | 80  | 84  |
| 14             |   | 2 | 7  | 11 | 16 | 21 | 26 | 31 | 36 | 41 | 46 | 51 | 56 | 61 | 66  | 71  | 77  | 82  | 87  | 92  |
| 15             |   | 3 | 7  | 12 | 18 | 23 | 28 | 33 | 39 | 44 | 50 | 55 | 61 | 66 | 72  | 77  | 83  | 88  | 94  | 100 |
| 16             |   | 3 | 8  | 14 | 19 | 25 | 30 | 36 | 42 | 48 | 54 | 60 | 65 | 71 | 77  | 83  | 89  | 95  | 101 | 107 |
| 17             |   | 3 | 9  | 15 | 20 | 26 | 33 | 39 | 45 | 51 | 57 | 64 | 70 | 77 | 83  | 89  | 96  | 102 | 109 | 115 |
| 18             |   | 4 | 9  | 16 | 22 | 28 | 35 | 41 | 48 | 55 | 61 | 68 | 75 | 82 | 88  | 95  | 102 | 109 | 116 | 123 |
| 19             | 0 | 4 | 10 | 17 | 23 | 30 | 37 | 44 | 51 | 58 | 65 | 72 | 80 | 87 | 94  | 101 | 109 | 116 | 123 | 130 |
| 20             | 0 | 4 | 11 | 18 | 25 | 32 | 39 | 47 | 54 | 62 | 69 | 77 | 84 | 92 | 100 | 107 | 115 | 123 | 130 | 138 |



**Probabilities associated with values as large as observed values of Kruskal-Wallis H Statistic.**  
**i.e.  $p_0 = P(H > H^*)$  where  $H^* = H_{cal}$ .**

| Sample sizes |       |       | H      | p    | Sample sizes |       |       | H      | p    |
|--------------|-------|-------|--------|------|--------------|-------|-------|--------|------|
| $n_1$        | $n_2$ | $n_3$ |        |      | $n_1$        | $n_2$ | $n_3$ |        |      |
| 2            | 1     | 1     | 2.7000 | .500 | 4            | 3     | 2     | 6.4444 | .008 |
|              |       |       |        |      |              |       |       | 6.3000 | .011 |
| 2            | 2     | 2     | 3.6000 | .200 |              |       |       | 5.4444 | .046 |
|              |       |       |        |      |              |       |       | 5.4000 | .051 |
| 2            | 2     | 2     | 4.5714 | .067 |              |       |       | 4.5111 | .098 |
|              |       |       | 3.7143 | .200 |              |       |       | 4.4444 | .102 |
| 3            | 1     | 1     | 3.2000 | .300 | 4            | 3     | 3     | 6.7455 | .010 |
|              |       |       |        |      |              |       |       | 6.7455 | .010 |
| 3            | 2     | 1     | 4.2857 | .100 |              |       |       | 6.7091 | .013 |
|              |       |       | 3.8571 | .100 |              |       |       | 5.7909 | .046 |
| 3            | 2     | 2     | 5.3572 | .029 |              |       |       | 4.7091 | .092 |
|              |       |       | 4.7143 | .048 |              |       |       | 4.7000 | .101 |
|              |       |       | 4.5000 | .067 |              |       |       |        |      |
|              |       |       | 4.4643 | .105 | 4            | 4     | 1     | 6.6667 | .010 |
|              |       |       |        |      |              |       |       | 6.1667 | .022 |
| 3            | 3     | 1     | 5.1429 | .043 |              |       |       | 4.9667 | .048 |
|              |       |       | 4.5714 | .100 |              |       |       | 4.8667 | .054 |
|              |       |       | 4.0000 | .129 |              |       |       | 4.1667 | .082 |
| 3            | 3     | 2     | 6.2500 | .011 |              |       |       | 4.0667 | .102 |
|              |       |       | 5.3611 | .032 | 4            | 4     | 2     | 7.0364 | .006 |
|              |       |       | 5.1389 | .061 |              |       |       | 6.8727 | .011 |
|              |       |       | 4.5556 | .100 |              |       |       | 5.4545 | .046 |
|              |       |       | 4.2500 | .121 |              |       |       | 5.2364 | .052 |
| 3            | 3     | 3     | 7.2000 | .004 |              |       |       | 4.5545 | .098 |
|              |       |       | 6.4889 | .011 |              |       |       | 4.4455 | .103 |
|              |       |       | 5.6889 | .029 | 4            | 4     | 3     | 7.1439 | .010 |
|              |       |       | 5.6000 | .050 |              |       |       | 7.1364 | .011 |
|              |       |       | 5.0667 | .086 |              |       |       | 5.5985 | .049 |
|              |       |       | 4.6222 | .100 |              |       |       | 5.5758 | .051 |
| 4            | 1     | 1     | 3.5714 | .200 |              |       |       | 4.5455 | .099 |
| 4            | 2     | 1     | 4.8214 | .057 |              |       |       | 4.4773 | .102 |
|              |       |       | 4.5000 | .076 | 4            | 4     | 4     | 7.6538 | .008 |
|              |       |       | 4.0179 | .114 |              |       |       | 7.5385 | .011 |
| 4            | 2     | 2     | 6.0000 | .014 |              |       |       | 5.6923 | .049 |
|              |       |       | 5.3333 | .033 |              |       |       | 5.6538 | .054 |
|              |       |       | 5.1250 | .052 |              |       |       | 4.6539 | .097 |
|              |       |       | 4.4583 | .100 |              |       |       | 4.5001 | .104 |
|              |       |       | 4.1887 | .105 | 5            | 1     | 1     | 3.8571 | .143 |
| 4            | 3     | 1     | 5.8333 | .021 | 5            | 2     | 1     | 5.2500 | .036 |
|              |       |       | 5.2083 | .060 |              |       |       | 5.0000 | .048 |
|              |       |       | 5.0000 | .057 |              |       |       | 4.4500 | .071 |
|              |       |       | 4.0556 | .093 |              |       |       | 4.2000 | .095 |



|   |   |   |        |      |   |   |   |        |      |
|---|---|---|--------|------|---|---|---|--------|------|
|   |   |   | 3.8889 | .129 |   |   |   | 4.0500 | .119 |
| 5 | 2 | 2 | 6.5333 | .008 | 5 | 4 | 4 | 7.7604 | .009 |
|   |   |   | 6.1333 | .013 |   |   |   | 7.7440 | .011 |
|   |   |   | 5.1600 | .034 |   |   |   | 5.6571 | .049 |
|   |   |   | 5.0400 | .056 |   |   |   | 5.6176 | .050 |
|   |   |   | 4.3733 | .090 |   |   |   | 4.6187 | .100 |
|   |   |   | 4.2933 | .122 |   |   |   | 4.5527 | .102 |
| 5 | 3 | 1 | 6.4000 | .012 | 5 | 5 | 1 | 7.3091 | .009 |
|   |   |   | 4.9600 | .048 |   |   |   | 6.8364 | .011 |
|   |   |   | 4.8711 | .052 |   |   |   | 5.1273 | .046 |
|   |   |   | 4.0178 | .095 |   |   |   | 4.9091 | .53  |
|   |   |   | 3.8400 | .123 |   |   |   | 4.1091 | .086 |
| 5 | 3 | 2 | 6.9091 | .009 |   |   |   | 4.0364 | .105 |
|   |   |   | 6.8281 | .010 | 5 | 5 | 2 | 7.3385 | .010 |
|   |   |   | 5.2509 | .049 |   |   |   | 7.2692 | .010 |
|   |   |   | 5.1055 | .052 |   |   |   | 5.3385 | .047 |
|   |   |   | 4.6509 | .091 |   |   |   | 5.2464 | .051 |
|   |   |   | 4.4945 | .101 |   |   |   | 4.6231 | .097 |
| 5 | 3 | 3 | 7.0788 | .009 |   |   |   | 4.5077 | .100 |
|   |   |   | 6.9818 | .011 | 5 | 5 | 3 | 7.5780 | .010 |
|   |   |   | 5.6465 | .049 |   |   |   | 7.5429 | .010 |
|   |   |   | 5.5152 | .051 |   |   |   | 5.7055 | .046 |
|   |   |   | 4.5333 | .097 |   |   |   | 5.6264 | .051 |
|   |   |   | 4.4121 | .109 |   |   |   | 4.5451 | .100 |
| 5 | 4 | 1 | 6.9545 | .008 |   |   |   | 4.5363 | .102 |
|   |   |   | 6.8400 | .011 | 5 | 5 | 4 | 7.8229 | .010 |
|   |   |   | 4.9855 | .044 |   |   |   | 7.7914 | .010 |
|   |   |   | 4.8600 | .056 |   |   |   | 5.6657 | .049 |
|   |   |   | 3.9873 | .098 |   |   |   | 5.6429 | .050 |
|   |   |   | 3.9600 | .102 |   |   |   | 4.5229 | .099 |
| 5 | 4 | 2 | 7.2045 | .009 |   |   |   | 4.5200 | .101 |
|   |   |   | 7.1182 | .010 | 5 | 5 | 5 | 8.0000 | .009 |
|   |   |   | 5.2727 | .049 |   |   |   | 7.9800 | .010 |
|   |   |   | 5.2682 | .050 |   |   |   | 5.7800 | .049 |
|   |   |   | 4.5409 | .098 |   |   |   | 5.6600 | .051 |
|   |   |   | 4.5182 | .101 |   |   |   | 4.5600 | .100 |
|   |   |   |        | .10  |   |   |   | 4.5000 | .102 |
| 5 | 4 | 3 | 7.4449 | .010 |   |   |   |        |      |
|   |   |   | 7.3949 | .011 |   |   |   |        |      |
|   |   |   | 5.6564 | .049 |   |   |   |        |      |
|   |   |   | 5.6308 | .050 |   |   |   |        |      |
|   |   |   | 4.5487 | .099 |   |   |   |        |      |
|   |   |   | 4.5231 | .103 |   |   |   |        |      |



**Probabilities associated with values as small as observed values of  $x$  in the binomial distribution with parameter  $n$  and  $p = \frac{1}{2}$**

$$\text{i.e. } p_0 = P(X \leq x) = \sum_{x=0}^{x=k} \binom{n}{x} \left(\frac{1}{2}\right)^n; k = 0, 1, 2, \dots (n-1).$$

| $n \backslash x$ | 0   | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  | 11  | 12  | 13  | 14  | 15  |
|------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 1                | 5   | *   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 2                | 250 | 500 | *   |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 3                | 125 | 500 | 875 | *   |     |     |     |     |     |     |     |     |     |     |     |     |
| 4                | 063 | 313 | 688 | 938 | *   |     |     |     |     |     |     |     |     |     |     |     |
| 5                | 031 | 188 | 500 | 812 | 969 | *   |     |     |     |     |     |     |     |     |     |     |
| 6                | 016 | 109 | 344 | 656 | 891 | 984 | *   |     |     |     |     |     |     |     |     |     |
| 7                | 008 | 062 | 227 | 500 | 773 | 938 | 992 | *   |     |     |     |     |     |     |     |     |
| 8                | 004 | 035 | 145 | 363 | 637 | 855 | 965 | 996 | *   |     |     |     |     |     |     |     |
| 9                | 002 | 020 | 090 | 254 | 500 | 746 | 910 | 980 | 998 | *   |     |     |     |     |     |     |
| 10               | 001 | 011 | 055 | 172 | 377 | 623 | 828 | 945 | 989 | 999 | *   |     |     |     |     |     |
| 11               |     | 006 | 033 | 113 | 274 | 500 | 726 | 887 | 967 | 994 | *   | *   |     |     |     |     |
| 12               |     | 003 | 019 | 073 | 194 | 387 | 613 | 806 | 927 | 981 | 997 | *   | *   |     |     |     |
| 13               |     | 002 | 011 | 046 | 133 | 291 | 500 | 709 | 867 | 954 | 989 | 998 | *   | *   |     |     |
| 14               |     | 001 | 006 | 029 | 090 | 212 | 395 | 605 | 788 | 910 | 971 | 994 | 999 | *   | *   |     |
| 15               |     |     | 004 | 018 | 059 | 151 | 304 | 500 | 696 | 849 | 941 | 982 | 996 | *   | *   | *   |
| 16               |     |     | 002 | 011 | 038 | 105 | 227 | 402 | 598 | 773 | 895 | 962 | 989 | 998 | *   | *   |
| 17               |     |     | 001 | 006 | 025 | 072 | 166 | 315 | 500 | 685 | 834 | 928 | 975 | 994 | 999 | *   |
| 18               |     |     | 001 | 004 | 015 | 048 | 119 | 240 | 407 | 593 | 760 | 881 | 952 | 985 | 996 | 999 |
| 19               |     |     |     | 002 | 010 | 032 | 084 | 180 | 324 | 500 | 676 | 820 | 916 | 968 | 990 | 998 |
| 20               |     |     |     | 001 | 006 | 021 | 058 | 132 | 252 | 412 | 588 | 748 | 868 | 942 | 979 | 994 |
| 21               |     |     |     | 001 | 004 | 013 | 039 | 095 | 192 | 332 | 500 | 668 | 808 | 905 | 961 | 987 |
| 22               |     |     |     |     | 002 | 008 | 026 | 067 | 143 | 262 | 416 | 584 | 738 | 857 | 933 | 974 |
| 23               |     |     |     |     | 001 | 005 | 017 | 047 | 105 | 202 | 339 | 500 | 661 | 798 | 895 | 953 |
| 24               |     |     |     |     | 001 | 003 | 011 | 032 | 076 | 154 | 271 | 419 | 581 | 729 | 846 | 924 |
| 25               |     |     |     |     | 002 | 007 | 022 | 054 | 115 | 212 | 345 | 500 | 655 | 788 | 885 |     |

Note: To save space decimal points are omitted in the p's.  
 \* = 1 or approximately 1.0